Conformal Quantile Nowcasting*

Martin Fankhauser†

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Abstract

Quantile nowcasts of GDP growth are central to monitoring macroeconomic tail risk but usually require computationally demanding models. We propose conformal quantile nowcasting: the quantile nowcast is a model's point nowcast plus an empirical quantile of its past real-time errors. The procedure works with any point nowcasting model, is calibrated in finite samples under exchangeability, never produces crossing quantiles and costs essentially nothing beyond the point nowcast. A conditional version replaces the empirical quantile by a quantile regression of past errors on financial conditions, so that the lower tail widens when risks rise. In weekly nowcasts of US GDP growth, conformal calibration around a Bayesian mixed-frequency model recovers about four fifths of the accuracy gain from stochastic volatility, and conditioning on financial conditions closes most of the remaining gap, at a small fraction of the computational cost. In simulations, the approach is robust to many regressors and outliers, which distort residual bootstraps.

Keywords: Conformal prediction, Nowcasting, Quantile forecasts, Growth-at-risk, Financial conditions, Stochastic volatility

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†Bocconi University, Milan, Italy. Email: martin.fankhauser@unibocconi.it

1 Introduction

Policy institutions increasingly monitor not only the most likely path of economic activity but also its downside risks. Following Adrian et al. (2019), the lower quantiles of the distribution of GDP growth, the *growth-at-risk*, have become a standard summary of these risks. Their central finding is that the lower quantiles of future growth move strongly with financial conditions, whereas the upper quantiles are stable, and a large literature has since forecast these quantiles with financial indicators (Giglio et al., 2016; Adams et al., 2021; Adrian et al., 2022; Brownlees and Souza, 2021; Plagborg-Møller et al., 2020). A more recent literature produces such quantile forecasts in real time from high-frequency data, so-called *quantile nowcasts* (Lima et al., 2020; Ferrara et al., 2022; Carriero et al., 2022). The leading approach combines a mixed-frequency model for the conditional mean with stochastic volatility. Carriero et al. (2022), henceforth CCM2022, show that a Bayesian mixed-frequency model with stochastic volatility (BMF-SV) delivers more accurate quantile nowcasts than the same model with constant volatility (BMF), mixed-data sampling (MIDAS) regressions and quantile regressions, although stochastic volatility does not improve the point nowcasts (see also Clark, 2011; Carriero et al., 2015, 2016). Stochastic volatility comes at a cost, however: many latent states must be estimated by simulation every time the nowcast is updated, and it is not obvious how to add it to models such as MIDAS regressions or machine-learning methods. Residual bootstraps for MIDAS models (Aastveit et al., 2017) are an alternative, but they re-estimate the model for every bootstrap replication and, as we show below, can be badly miscalibrated when the model contains many regressors.

This paper proposes a simple alternative, which we call *conformal quantile nowcasting*. Given any point nowcasting model, the α-quantile nowcast is the current point nowcast plus the empirical α-quantile of the model's past real-time nowcast errors (Algorithm 1). The procedure uses only the point nowcasts that are produced anyway, it can be wrapped around any model, and quantile nowcasts at different levels cannot cross. Constructing intervals from past forecast errors has a long tradition in applied forecasting (Williams and Goodman, 1971;

Lee and Scholtes, 2014; Knüppel, 2014). We show that, when applied to *signed* errors and one quantile level at a time, this practice is a one-sided version of split conformal prediction (Vovk et al., 2005; Lei et al., 2018). The connection supplies what the heuristic lacks: explicit conditions under which the quantile nowcasts are calibrated, and a precise statement of what they target.

What they target is the *marginal* quantile: the shift added to the point nowcast does not depend on the state of the economy. This is exactly where growth-at-risk is informative. We therefore extend the procedure with the conformal method with conditional guarantees of Gibbs et al. (2025). The empirical quantile of past errors is replaced by a quantile regression of past errors on observed variables, here the Chicago Fed's National Financial Conditions Index (NFCI). The resulting *conditional* conformal quantile nowcast is a growth-at-risk regression for nowcast errors: the point nowcasting model describes the conditional mean, and the quantile regression lets the lower tail widen when financial conditions tighten. In its finite-sample form, it is calibrated not only on average but under every reweighting of the data by the conditioning functions, for instance separately in periods of tight and loose financial conditions.

The paper makes four contributions. First, we formalize conformal quantile nowcasting for a real-time environment with publication lags and multiple quantile levels and state the properties it inherits from conformal prediction (Section 2). If the nowcast errors are exchangeable, the probability that the outcome falls below the α-quantile nowcast is within $2/(n+1)$ of α, where n is the number of past errors, and quantile nowcasts at different levels never cross. For weakly dependent (β-mixing) data, the results of Oliveira et al. (2024) imply that these guarantees hold approximately. Using signed rather than absolute errors matters, because standard two-sided conformal intervals control only the total miscoverage and can misallocate it between the tails when errors are skewed (Linusson et al., 2014).

Second, we adapt conditional conformal prediction to quantile nowcasting and link it to growth-at-risk (Section 2.3). We state the finite-sample guarantee of the exact version

and show that, with the short samples typical of macroeconomics, it becomes uninformative precisely in crises, when financial conditions reach values not seen in the past. We therefore use a plug-in version with restrictions motivated by the growth-at-risk evidence and by the small number of past errors, and we show how the results change when each restriction is removed.

Third, we compare conformal quantile forecasts with residual bootstraps for autoregressions (Pan and Politis, 2016) in Monte Carlo simulations (Section 3). Without exogenous regressors, both approaches are close to nominal coverage, with the bootstrap more accurate in very small samples and for extreme quantiles; the small-sample behavior of the conformal quantiles is exactly what the finite-sample theory predicts. With many exogenous regressors, the non-studentized bootstrap overcovers, while conformal quantiles are only mildly affected. The studentized bootstrap repairs the dimensionality problem but overcovers when the sample contains an outlier, whereas conformal quantiles, which depend on the errors only through an order statistic, are unaffected by the size of the outlier.

Fourth, we evaluate the approach in two real-time applications. In the main application (Section 4), we nowcast the 5% quantile of US GDP growth in each of 15 weeks per quarter in the setting of CCM2022, with 1989Q4–2019Q4 as evaluation sample. The results decompose the advantage of stochastic volatility. With constant volatility, the predictive 5% quantile of the BMF has a quantile score 20% higher than that of the BMF-SV. Applying Algorithm 1 to the point nowcast of the same BMF reduces the gap to 4% on average, so the conformal step recovers about four fifths of the gain from stochastic volatility without estimating any volatility process. The remaining gap is concentrated in recessions, when a constant shift cannot widen. Conditioning on the NFCI narrows the recession gap by about half, at almost no cost in expansions, and the conditional conformal quantile nowcast matches the BMF-SV on average (relative quantile score 1.003). It has lower scores than the BMF-SV early in the quarter and since 2000, and is never significantly worse. The point nowcast matters as much as the calibration step. Wrapped around a MIDAS regression, whose point nowcasts are less

accurate, conformal quantile nowcasts are better calibrated than MIDAS bootstraps but have scores about 15% higher than those of the BMF-SV, and conditioning on the NFCI does not help. In a second application (Section 5), we forecast all percentiles of daily Microsoft stock returns and compare with the dynamic multiple quantile (DMQ) model of Catania and Luati (2023). A conformalized autoregression yields quantile losses that are less than 1% higher than those of the DMQ in the center of the distribution and up to 6.5% higher in the tails, while it avoids quantile crossing by construction.

The paper relates to several strands of the literature. Conformal prediction converts point predictions into prediction sets with finite-sample coverage guarantees under exchangeability (Vovk et al., 2005; Lei et al., 2018); see Angelopoulos and Bates (2023) and Fontana et al. (2023) for reviews. One-sided conformal sets based on signed scores were introduced by Linusson et al. (2014) to control the coverage of each tail, and conformalized quantile regression (Romano et al., 2019) and distributional conformal prediction (Chernozhukov et al., 2021a) calibrate quantile or distribution models. A growing literature extends conformal prediction to dependent data (Chernozhukov et al., 2018; Gibbs and Candes, 2021; Gibbs and Candès, 2024; Barber et al., 2023; Xu and Xie, 2023; Oliveira et al., 2024) and to conditional guarantees (Gibbs et al., 2025). We contribute to the literature on density and quantile nowcasting (Aastveit et al., 2014, 2017; Lima et al., 2020; Ferrara et al., 2022; Carriero et al., 2022) a model-agnostic and computationally light way to obtain calibrated quantile nowcasts. Finally, our results speak to the debate on growth-at-risk. Adrian et al. (2019) and Adams et al. (2021) find that financial conditions predict the left tail of GDP growth, whereas Plagborg-Møller et al. (2020) find that financial variables add little beyond the conditional mean and provide no robust early warning of tail risks, and Brownlees and Souza (2021) find that a GARCH model forecasts growth-at-risk as accurately as quantile regressions on financial conditions. In our nowcasting setting, a volatility model and a quantile regression on financial conditions perform equally well on average, the gains from financial conditions are concentrated in the first half of the quarter, before much information on the current quarter

has arrived, and they stem entirely from the financial crisis of 2008–2009.

The remainder of the paper is organized as follows. Section 2 presents the method, its finite-sample properties, the conditional extension and the relation to conformal methods for dependent data. Section 3 reports the Monte Carlo evidence, Section 4 the nowcasting application and Section 5 the multiple quantile application. Section 6 concludes. Proofs are collected in Appendix A.

2 Conformal quantile nowcasts

2.1 Setting and algorithm

Let $y_t \in \mathbb{R}$ denote the target variable in period t, for instance quarterly GDP growth, and let $\mathcal{I}_t$ denote the information set available when the nowcast of y_t is made. In nowcasting, $\mathcal{I}_t$ contains past values of y that have already been published and high-frequency indicators released up to the nowcast date. A point nowcasting method $\mathcal{A}$, for instance a MIDAS regression, a Bayesian mixed-frequency model or a machine-learning algorithm, is estimated on $\mathcal{I}_t$ and produces the point nowcast $\hat{y}_t$. We place no restriction on $\mathcal{A}$; in particular, it can be re-estimated in every period.

Once y_t is published, the *signed* real-time nowcast error

$$e_t = y_t - \hat{y}_t \tag{1}$$

becomes available. Because of publication lags, the error for period s is known at the time of the nowcast for period t only if $s \leq t - \ell$, where $\ell \geq 1$ is the publication lag measured in periods of the target variable. For a window length $w \in \mathbb{N} \cup \{\infty\}$, the set of past errors available when nowcasting y_t is

$$\mathcal{E}_t = \{e_s : t - \ell - w < s \leq t - \ell,\ s \geq s_0\}, \qquad n_t = |\mathcal{E}_t|, \tag{2}$$

where s_0 is the first period for which an error is recorded and $w = \infty$ corresponds to an expanding window.

For a finite set $\mathcal{E}$ of n numbers with order statistics $e_{(1)} \leq \cdots \leq e_{(n)}$, let $\hat{Q}_\alpha(\mathcal{E})$ denote its empirical α-quantile. We consider two versions. The *conservative* version sets $\hat{Q}_\alpha(\mathcal{E}) = e_{(\lfloor \alpha(n+1) \rfloor)}$, with the convention $e_{(0)} = -\infty$. The *interpolated* version, which is the default in R (type 7 in Hyndman and Fan, 1996) and is used in all computations of this paper, sets $\hat{Q}_\alpha(\mathcal{E}) = e_{(j)} + \gamma(e_{(j+1)} - e_{(j)})$ with $h = (n-1)\alpha + 1$, $j = \lfloor h \rfloor$ and $\gamma = h - j$.

Definition 1 (Conformal quantile nowcast)**.** *The conformal α-quantile nowcast of y_t is*

$$\hat{q}_t(\alpha) = \hat{y}_t + \hat{Q}_\alpha(\mathcal{E}_t). \tag{3}$$

Algorithm 1 summarizes the procedure. In the empirical applications, ℓ is the number of periods until the outcome of period s is published: $\ell = 1$ for daily stock returns, and $\ell \in \{1, 2\}$ in the GDP application, depending on the week of the nowcasting cycle (see Section 4). The window is either expanding ($w = \infty$) or rolling.

Algorithm 1 Conformal quantile nowcasting

Input: quantile levels $0 < \alpha_1 < \cdots < \alpha_K < 1$; point nowcasting method $\mathcal{A}$; publication lag $\ell \geq 1$; window length w; first error period s_0 and first forecast period $t_0 > s_0$

for $t = s_0, s_0 + 1, \ldots, T$ **do**
  Estimate $\mathcal{A}$ on the information set $\mathcal{I}_t$ and compute the point nowcast $\hat{y}_t$
  if $t \geq t_0$ **then**
    Collect the published errors $\mathcal{E}_t = \{e_s : t - \ell - w < s \leq t - \ell,\ s \geq s_0\}$
    for $k = 1, \ldots, K$ **do**
      $\hat{q}_t(\alpha_k) \leftarrow \hat{y}_t + \hat{Q}_{\alpha_k}(\mathcal{E}_t)$ ▷ empirical quantile of signed errors
    end for
  end if
  Once y_t is published, record the signed error $e_t = y_t - \hat{y}_t$
end for

Output: quantile nowcasts $\hat{q}_t(\alpha_1) \leq \cdots \leq \hat{q}_t(\alpha_K)$ for $t = t_0, \ldots, T$

We refer to Algorithm 1 as CP1 and add the name of the point nowcasting method where needed, for instance CP1-MIDAS. As a benchmark we also consider the lower end of the standard two-sided split conformal interval, which uses absolute errors as conformity scores,

$\hat{q}_t^{\mathrm{CP2}}(\alpha) = \hat{y}_t - \hat{Q}_{1-2\alpha}(\{|e_s| : e_s \in \mathcal{E}_t\})$; we refer to this version as CP2. The computational cost of CP1 is that of the point nowcasts plus sorting at most n_t numbers per period. No additional model has to be estimated, and the point nowcasting model does not have to be re-estimated on resampled data as in the bootstrap.

2.2 Finite-sample properties

The conformal quantile nowcast is the lower end of a one-sided split conformal prediction set based on the signed score $y - \hat{y}$ (Linusson et al., 2014). Its properties therefore follow from standard conformal arguments (Vovk et al., 2005; Lei et al., 2018; Romano et al., 2019), which we restate for the quantile nowcast in (3).

Proposition 1 (Finite-sample calibration)**.** *Fix a forecast period t and suppose that the $n+1$ errors in $\mathcal{E}_t \cup \{e_t\}$ are exchangeable and almost surely distinct, where $n = n_t$. Let $e_{(1)} < \cdots < e_{(n)}$ denote the order statistics of $\mathcal{E}_t$.*

(i) For every $k \in \{1, \ldots, n\}$, $P\big(y_t \le \hat{y}_t + e_{(k)}\big) = k/(n+1)$.

(ii) For the conservative version of $\hat{Q}_\alpha$ and $\alpha \ge 1/(n+1)$,

$$\alpha - \frac{1}{n+1} < P\big(y_t \le \hat{q}_t(\alpha)\big) \le \alpha.$$

(iii) For the interpolated version of $\hat{Q}_\alpha$, with $j = \lfloor (n-1)\alpha \rfloor + 1$,

$$\frac{j}{n+1} \le P\big(y_t \le \hat{q}_t(\alpha)\big) \le \frac{j+1}{n+1}, \qquad \textit{so that} \qquad \Big|P\big(y_t \le \hat{q}_t(\alpha)\big) - \alpha\Big| \le \frac{2}{n+1}.$$

(iv) For both versions and any $\alpha < \alpha'$, $\hat{q}_t(\alpha) \le \hat{q}_t(\alpha')$ for every realization of the data.

The proof is given in Appendix A. Parts (i)–(iii) say that the conformal quantile nowcast is calibrated in finite samples: the frequency with which the outcome falls below $\hat{q}_t(\alpha)$ is within a known distance of α, without any assumption on the distribution of the errors or

on the point nowcasting method. Part (iv) says that quantile nowcasts at different levels never cross, because they share the same point nowcast and differ only through the monotone empirical quantile function. Four remarks clarify the scope of these results.

Remark 1 (When are the errors exchangeable?). Exchangeability holds, for instance, if the data are independent and identically distributed and the point nowcasting method is estimated once on a separate training sample, which is the classical split conformal setting (Vovk et al., 2005; Lei et al., 2018). Full conformal prediction preserves exchangeability for methods that are re-estimated on the augmented sample, provided the estimator treats the observations symmetrically, which rules out, for instance, estimators that weight observations by their position in time (Vovk et al., 2005; Barber et al., 2023). In nowcasting, neither condition holds exactly: the data are serially dependent and the point nowcasting model is re-estimated every period on a growing information set. Section 2.5 discusses the approximate validity of the procedure in this case, and Sections 3–5 assess it numerically.

Remark 2 (What is being forecast?). Proposition 1 concerns the *marginal* probability $P(y_t \leq \hat{q}_t(\alpha))$, averaged over the predictors. Conditional calibration, $P(y_t \leq \hat{q}_t(\alpha) \mid x_t) = \alpha$ for all values of the predictors x_t, cannot be guaranteed in finite samples without further assumptions (Lei and Wasserman, 2014). Suppose $y_t = \mu(x_t) + \varepsilon_t$ with ε_t independent of x_t, and that the point nowcast is consistent for $\mu(x_t)$. Then the conformal quantile nowcast converges to $\mu(x_t) + Q_\alpha(\varepsilon_t)$, the conditional α-quantile of y_t. If instead the dispersion of ε_t varies with x_t or over time, the conformal quantile nowcast remains marginally calibrated but its shape around the point nowcast does not adapt to the conditional dispersion. This is precisely the feature that stochastic volatility and quantile regression models are designed to capture. Section 2.3 extends the procedure so that the shift can vary with observed indicators, and Section 4 quantifies how much is lost by relying on marginal calibration around a good point nowcast.

Remark 3 (Number of errors and multiple quantiles). By part (ii), the conservative version can only target levels $\alpha \geq 1/(n+1)$: with fewer than $1/\alpha - 1$ past errors, a calibrated

α-quantile nowcast does not exist without further assumptions. The interpolated version always returns a finite value, but by part (iii) its exceedance probability is at least $1/(n+1)$. The same logic applies to multiple quantiles: to obtain distinct forecasts of all percentiles, at least 100 past errors are required.

Remark 4 (One-sided versus two-sided scores). Standard conformal intervals use absolute errors $|e_s|$ as conformity scores. A $(1-2\alpha)$ interval then guarantees that the *total* probability outside the interval is approximately 2α, but it does not control how this probability is split between the two tails. If the errors are skewed, the probability below the lower end can be anywhere between 0 and 2α. Signed errors guarantee calibration for each tail separately (Linusson et al., 2014). The cost is that a one-sided interval with the same nominal level is wider. This cost is immaterial when the object of interest is a quantile.

2.3 Conditional conformal quantile nowcasts

Remark 2 identifies the main limitation of Algorithm 1: the shift $\hat{Q}_\alpha(\mathcal{E}_t)$ added to the point nowcast does not depend on the state of the economy. If downside risk rises in some states, for instance when financial conditions tighten, the conformal quantile nowcast is too high in those states and too low in the others, while remaining calibrated on average. The growth-at-risk literature suggests that this is relevant for GDP growth: Adrian et al. (2019) show that the lower quantiles of future growth fall sharply when financial conditions tighten, whereas the upper quantiles hardly move. Exact conditional calibration given all predictors is impossible in finite samples without further assumptions (Lei and Wasserman, 2014). Gibbs et al. (2025) propose an intermediate target: calibration over a chosen class of functions of observed variables. We adapt their method to quantile nowcasting.

Let z_t denote a small vector of variables observed at the nowcast date, for instance a financial conditions index, and let $\phi(z) = (1, \phi_2(z), \ldots, \phi_d(z))'$ be a vector of d features whose first element is a constant. Each past error $e_s \in \mathcal{E}_t$ is paired with the features z_s observed when the nowcast of period s was made. Let $\rho_\alpha(u) = u\,(\alpha - \mathbf{1}\{u < 0\})$ denote the check

function of quantile regression (Koenker and Bassett, 1978).

Definition 2 (Conditional conformal quantile nowcast)**.** *The (plug-in) conditional conformal α-quantile nowcast of y_t is*

$$\hat{q}_t^c(\alpha) = \hat{y}_t + \phi(z_t)'\hat{\beta}_t, \qquad \hat{\beta}_t \in \arg\min_{\beta\in\mathbb{R}^d} \sum_{e_s\in\mathcal{E}_t} \rho_\alpha\big(e_s - \phi(z_s)'\beta\big). \tag{4}$$

Its finite-sample version adds the target period to the quantile regression with an imputed error E,

$$\begin{aligned} \tilde{q}_t^c(\alpha) &= \hat{y}_t + \inf\big\{E\in\mathbb{R} : E \geq \phi(z_t)'\hat{\beta}_t(E)\big\}, \\ \hat{\beta}_t(E) &\in \arg\min_{\beta\in\mathbb{R}^d} \Big\{ \sum_{e_s\in\mathcal{E}_t} \rho_\alpha\big(e_s - \phi(z_s)'\beta\big) + \rho_\alpha\big(E - \phi(z_t)'\beta\big)\Big\}. \end{aligned} \tag{5}$$

With a constant only, $\phi(z) = 1$, the quantile regression in (4) returns an empirical α-quantile of the past errors and $\hat{q}_t^c(\alpha)$ reduces to Algorithm 1, while $\tilde{q}_t^c(\alpha)$ reduces to its conservative version. With additional features, the shift added to the point nowcast varies with z_t. We refer to the conditional conformal quantile nowcast as CCP, again adding the name of the point nowcasting method where needed. The following result adapts the main guarantee of Gibbs et al. (2025) to the present setting.

Proposition 2 (Conditional calibration)**.** *Fix a forecast period t and suppose that the $n+1$ pairs $\{(z_s, e_s) : e_s \in \mathcal{E}_t\} \cup \{(z_t, e_t)\}$ are exchangeable, where $n = n_t$, and that $\hat{\beta}_t(E)$ is a symmetric function of the $n+1$ observations in (5). Then, for every function $f(z) = \phi(z)'b$, $b \in \mathbb{R}^d$, with $f \geq 0$ and $\mathbb{E}[f(z_t)] > 0$,*

$$P_f\big(y_t < \tilde{q}_t^c(\alpha)\big) \leq \alpha, \qquad \textit{where} \quad P_f(A) = \frac{\mathbb{E}\big[f(z_t)\,\mathbf{1}_A\big]}{\mathbb{E}[f(z_t)]}.$$

The proof is given in Appendix A. P_f reweights the distribution of the data by $f(z_t)$, so Proposition 2 states that the quantile nowcast is calibrated, in the sense of not exceeding the nominal miss probability, under every reweighting in the class spanned by the features. Three special cases illustrate this. With $f = 1$ we recover marginal calibration as in Proposition 1(ii).

With $\phi(z) = (1, \mathbf{1}\{z \in G\})'$ for a set G, both $\mathbf{1}\{z \in G\}$ and $1 - \mathbf{1}\{z \in G\}$ belong to the class, so that $P(y_t < \tilde{q}_t^c \mid z_t \in G) \leq \alpha$ and $P(y_t < \tilde{q}_t^c \mid z_t \notin G) \leq \alpha$: the quantile nowcast is calibrated separately in, for instance, periods of tight and loose financial conditions, as in Mondrian conformal prediction (Vovk et al., 2005). With $\phi(z) = (1, z^+)'$, where $z^+ = \max(z, 0)$, calibration holds under every reweighting $1 + c\, z^+$, $c \geq 0$, that puts more weight on periods in which financial conditions are tighter than average. Gibbs et al. (2025) also show that the miss probability is not much below α: the slack is of order $d/(n+1)$ and vanishes with randomization.

Three practical issues arise in macroeconomic applications, where the number of past errors is small.

Remark 5 (Extrapolation and the finite-sample version). The finite-sample version (5) protects against extrapolation. If $\phi(z_t)$ lies outside the range of the features observed in the past, the quantile regression can fit the target period exactly for any imputed error E, and the lower bound can become $-\infty$. This is the correct answer to the question it addresses, because the past errors contain no information about the tail at unprecedented values of z_t, but it occurs precisely in crises, when a quantile nowcast is most needed. In our application, the finite-sample version is unbounded in 41 of the 1,815 nowcasts of the evaluation sample, all between 2008Q1 and 2009Q2, and in 35 of them the NFCI exceeds every value in the calibration sample. The plug-in version (4) instead extrapolates linearly. It relates to the finite-sample version as the interpolated empirical quantile relates to the conservative one in Algorithm 1, and we use it in the applications.

Remark 6 (Restrictions). With few past errors, the quantile regression is poorly estimated: at level α, roughly $1/\alpha$ observations are needed to identify each coefficient from the tail. We therefore impose two restrictions, motivated by this sample-size argument and by the growth-at-risk evidence. First, we estimate the slopes only once at least d/α periods of past errors are available and use Algorithm 1 before. Second, we restrict the sign of the slopes: if $\phi_j(z)$ measures the tightness of financial conditions, its coefficient is restricted to

be non-positive, so that tighter conditions can only lower the quantile nowcast. With a single slope, the restricted estimator sets the slope to zero whenever the unrestricted estimate has the wrong sign, and the nowcast then coincides with Algorithm 1. Both restrictions change the class of functions in Proposition 2 in a data-dependent way, so the proposition does not apply to the restricted plug-in version. We report its performance with and without each restriction in Section 4.

Remark 7 (Multiple quantiles). Unlike Algorithm 1, quantile regressions estimated separately at different levels can cross, so Proposition 1(iv) does not carry over. Monotonicity can be restored by sorting the quantile nowcasts at each date, which does not increase the quantile loss (Chernozhukov et al., 2010). Our nowcasting application targets a single level.

Relation to growth-at-risk. Adrian et al. (2019) estimate quantile regressions of future GDP growth on current growth and the NFCI. The conditional conformal quantile nowcast (4) is a quantile regression of the *nowcast errors* on the NFCI. The division of labor differs: the point nowcasting model, which can contain many indicators and be estimated by any method, describes the conditional mean, and the quantile regression, which contains only a constant and a few risk indicators, describes how the lower tail around the point nowcast changes with the risk indicators. If the point nowcast already captures the effect of financial conditions on the mean, the slope in (4) measures their effect on the remaining downside risk, that is, the asymmetry emphasized by Adrian et al. (2019). The positive part z^+ allows only tighter-than-average financial conditions to matter, in line with their finding that loose financial conditions do not raise the upper quantiles. The approach also differs from conformalized quantile regression (Romano et al., 2019), which estimates a rich quantile model and calibrates it with a constant shift. Here the point model is rich and the quantile model is sparse.

2.4 Relation to other conformal approaches to quantiles

The idea of constructing intervals from the empirical distribution of past forecast errors is old (Williams and Goodman, 1971) and is still widely used, for instance to construct fan charts (Lee and Scholtes, 2014; Knüppel, 2014). Algorithm 1 differs from these approaches in two respects. It treats each tail separately through signed errors, and it computes the errors in real time, so that each error reflects the information and the estimated model available when the corresponding nowcast was made. Viewing the procedure as conformal prediction provides the finite-sample guarantees of Proposition 1 and the results for dependent data discussed below.

Within the conformal literature, two approaches produce quantile forecasts by construction. Conformalized quantile regression (Romano et al., 2019) adjusts the output of a quantile regression, and distributional conformal prediction (DCP; Chernozhukov et al., 2021a) conformalizes an estimated conditional distribution, for instance the posterior predictive distribution of a Bayesian model. Both approaches use quantile or distribution estimates as inputs and can therefore adapt to conditional heteroskedasticity, but they require such estimates to be available. Our approach targets the opposite situation, in which accurate point nowcasts exist but quantile or density models are costly to estimate. In the nowcasting application we include a one-sided version of DCP applied to the posterior predictive distribution of the BMF as a benchmark (DCP-BMF), which is valid for strongly mixing processes and also rules out quantile crossing (Chernozhukov et al., 2021a). More generally, Vovk et al. (2019) show that the collection of one-sided conformal sets at all levels forms a predictive distribution; Algorithm 1 with many quantile levels is a split-conformal instance of this idea. Locally weighted conformal prediction (Lei et al., 2018) instead scales the errors by an estimate of their dispersion. We use this variant in Section 5, where a volatility forecast is readily available. Conformalizing Bayesian predictive distributions can also be viewed as a protective layer against misspecification (Fong and Holmes, 2021), a direction we do not pursue further.

2.5 Beyond exchangeability

Time series data are not exchangeable, and a large literature extends conformal prediction to this case. These extensions typically require additional assumptions on the data generating process, deliver weaker guarantees, or both. We summarize the main approaches and their implications for quantile nowcasting.

For weakly dependent data, Oliveira et al. (2024) show that split conformal prediction remains approximately valid for stationary β-mixing processes, a class that includes, for instance, stationary ARMA and GARCH processes under standard conditions. The coverage penalty depends on the mixing coefficients and vanishes as the number of calibration scores grows. Their results hold for any conformity score, so they apply to the signed errors used in Algorithm 1. They are stated for a model estimated once on a training sample, and re-estimating the model as new data arrive is discussed only informally. Because these results require no change to the algorithm, they justify applying Algorithm 1 directly to several quantile levels, which preserves the monotonicity in Proposition 1(iv). We follow this route and assess its performance numerically.

A second strand builds on Gibbs and Candes (2021). Adaptive conformal inference (ACI) treats the level of the empirical quantile as a parameter that is updated online: the level is moved outward after a miss and inward otherwise. Instead of modeling distribution shifts in the high-dimensional joint distribution of the data (as, e.g., Cauchois et al., 2023), the problem reduces to tracking a scalar. The key tuning parameter is the step size of the update. Gibbs and Candès (2024) aggregate over a grid of step sizes with exponential reweighting (fully adaptive conformal inference, FACI) and obtain coverage guarantees over arbitrary time intervals. Zaffran et al. (2022) and Bastani et al. (2022) provide related proposals with long-run guarantees, Feldman et al. (2023) extend the idea to risk control, and Angelopoulos et al. (2023) and Xu and Xie (2023) additionally forecast the quantile to be used in the next period. These methods require no assumptions on the data generating process, but the levels are updated separately for each quantile, so monotonicity across quantile levels is not

guaranteed. We include FACI wrapped around CP1 as a benchmark.

A third strand reweights the conformity scores. Tibshirani et al. (2019) use a likelihood ratio to accommodate covariate shift, and Barber et al. (2023) use fixed, data-independent weights that accommodate gradual distribution shifts. Their guarantees deteriorate with the size of the violation of exchangeability. Barber et al. (2023) also handle non-symmetric estimators by randomly swapping the test point with a calibration point. In autoregressive models this swapping can harm predictive accuracy, because recent observations are the most informative. The sequential approach of Xu and Xie (2023) can be viewed as a data-driven choice of such weights. A different line of work builds on block permutations of conformity scores (Chernozhukov et al., 2018, 2021b). Xu and Xie (2021) use this idea for a bootstrap aggregation scheme related to the jackknife+ (Barber et al., 2021), and Jensen et al. (2022) extend conformalized quantile regression to time series. Finally, Stankeviciute et al. (2021) and Lin et al. (2022) consider multi-horizon and cross-sectional time series, respectively.

A fourth strand, discussed in Section 2.3, targets calibration conditional on observed variables rather than robustness to unobserved distribution shifts (Gibbs et al., 2025). Based on the results of Oliveira et al. (2024) and on its performance relative to these alternatives in our applications, we use Algorithm 1 and its conditional extension as the main methods. As benchmarks, we report FACI wrapped around CP1, because it requires no assumptions on the data generating process, and, in the nowcasting application, DCP applied to the BMF. All approaches above can in principle be used to produce quantile forecasts, but only some of them guarantee non-crossing quantiles.

3 Monte Carlo simulations

This section compares conformal quantile forecasts with bootstrap quantile forecasts for linear autoregressions, with and without exogenous regressors, and compares conformal and DMQ forecasts of multiple quantiles. The designs mirror the two empirical applications.

For the purely autoregressive case, we compare with the forward bootstrap of Pan and Politis (2016). For the models with exogenous regressors, we apply the same procedure and treat the exogenous regressors as fixed, as is common in the forecasting literature (e.g., Aastveit et al., 2017). We consider the non-studentized forward bootstrap with predictive residuals (FpB) and its studentized counterpart with fitted residuals (FsB), Algorithms 3.2 and 3.3 in Pan and Politis (2016).[1] CP1 denotes Algorithm 1 with an expanding window of one-step-ahead out-of-sample errors from the re-estimated autoregression, and CP2 the lower end of the corresponding two-sided interval. In each replication, a sample of size T is generated and used to construct the forecast for period $T+1$. To measure coverage conditional on the sample, we draw 1,000 realizations of y_{T+1} for each simulated sample and record the fraction that falls below the forecast quantile. We report averages over 1,000 replications.[2]

In the first design, we follow Pan and Politis (2016) and simulate from an AR(1) model with coefficient $\phi = 0.5$, with errors drawn either from a standard normal distribution or from a χ^2 distribution with 5 degrees of freedom, shifted to have mean zero so that the model for the conditional mean is correctly specified.[3] Errors for CP1 are collected from period 20 onward, so CP1 is based on $n = T - 20$ errors.

The top panel of Table 1 reports the exceedance frequencies for the 1%, 5% and 10% quantiles with $T = 50$, 100 and 200 and normal errors; the last three rows report the coverage of two-sided prediction intervals with nominal level $1 - 2\alpha$. The bootstraps are closer to the nominal level than the conformal methods for small samples and extreme quantiles, in line with the simulation results of Wang and Politis (2021). This small-sample behavior of CP1 is

[1] We do not consider the studentized version with predictive residuals because Pan and Politis (2016) report a similar performance to the two versions considered.

[2] The bootstrap uses 1,000 bootstrap replications. Conformal prediction guarantees marginal rather than conditional coverage without further assumptions on the data generating process (Lei and Wasserman, 2014); see also Angelopoulos and Bates (2023). Under additional assumptions some notion of conditional coverage can be achieved (Oliveira et al., 2024; Chernozhukov et al., 2021a), which is why Wang and Politis (2021) focus their comparison of the model-free bootstrap on DCP.

[3] Oliveira et al. (2024) show in simulations that the split conformal algorithm works well even for highly persistent processes. In our simulations, the autoregressive parameter was not crucial as long as the process remains stationary.

the one predicted by Proposition 1. For $T = 50$, CP1 uses $n = 30$ errors, and part (iii) of the proposition implies an exceedance probability between $1/31 \approx 0.032$ and $2/31 \approx 0.065$ for $\alpha = 0.01$; the simulated value is 0.036. In fact, all nine CP1 entries in the top panel lie within the bounds of part (iii), even though the re-estimated errors are only approximately exchangeable. The deviation from the nominal level for $\alpha = 0.01$ in small samples is therefore a consequence of the small number of past errors (Remark 3) and vanishes as T grows.

Table 1: Monte Carlo simulation: AR(1)

	Standard normal errors $\epsilon_t \sim N(0,1)$								
	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$
CP1	0.036	0.020	0.015	0.074	0.060	0.054	0.122	0.108	0.103
CP2	0.022	0.014	0.012	0.059	0.053	0.051	0.105	0.101	0.099
FpB	0.016	0.014	0.013	0.051	0.052	0.052	0.100	0.100	0.101
FsB	0.015	0.013	0.013	0.051	0.052	0.052	0.101	0.101	0.102
PI-CP	0.957	0.971	0.976	0.882	0.895	0.899	0.791	0.799	0.800
PI-FpB	0.968	0.973	0.975	0.898	0.896	0.897	0.803	0.800	0.798
PI-FsB	0.971	0.974	0.975	0.898	0.896	0.898	0.801	0.799	0.798
	Shifted χ^2 errors $\epsilon_t \sim \chi^2(5)$								
	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$
CP1	0.038	0.015	0.011	0.075	0.053	0.050	0.121	0.103	0.099
CP2	0.002	0.000	0.000	0.025	0.012	0.009	0.082	0.071	0.071
FpB	0.007	0.006	0.008	0.041	0.043	0.047	0.092	0.093	0.097
FsB	0.007	0.006	0.008	0.044	0.045	0.048	0.097	0.096	0.098
PI-CP	0.957	0.971	0.976	0.883	0.898	0.903	0.789	0.803	0.805
PI-FpB	0.975	0.979	0.979	0.903	0.903	0.902	0.806	0.804	0.803
PI-FsB	0.976	0.981	0.980	0.903	0.904	0.901	0.803	0.804	0.801

Note: Results from 1,000 replications of $y_t = 0.5y_{t-1} + \epsilon_t$ with $\epsilon_t \sim N(0,1)$ (top panel) and ϵ_t drawn from a $\chi^2(5)$ distribution shifted such that $\mathbb{E}[\epsilon_t] = 0$ (bottom panel). The column header n denotes the sample size used for estimation (T in the text). Exceedance frequencies are computed from 1,000 draws of y_{n+1} for each replication and averaged across replications. CP1: one-sided conformal quantile (Algorithm 1); CP2: lower end of the two-sided conformal interval; FpB: non-studentized forward bootstrap with predictive residuals (Algorithm 3.2 in Pan and Politis (2016)); FsB: studentized forward bootstrap with fitted residuals (Algorithm 3.3 in Pan and Politis (2016)). The last three rows report the coverage of two-sided prediction intervals with nominal level $1 - 2\alpha$ obtained from two-sided conformal prediction and the two bootstraps.

The bottom panel repeats the exercise with skewed χ^2 errors, so that the point forecast makes asymmetric mistakes. CP1 performs as with normal errors, whereas CP2 is heavily miscalibrated: its exceedance frequency is at most 0.002 for $\alpha = 0.01$ and between 0.009 and 0.025 for $\alpha = 0.05$. This is the misallocation of tail probabilities described in Remark 4. The bootstraps again perform best, relative to the conformal methods, in small samples and for extreme quantiles.

The second design, reported in Table 2, augments the autoregression with $K = 10$, 25 and 50 exogenous regressors drawn from a standard normal distribution, with $T = 100$, so that the ratio of regressors to observations is $\kappa = K/T = 0.10$, 0.25 and 0.50.[4] This design mimics a MIDAS regression with many high-frequency regressors and a lagged dependent variable, as bootstrapped by Aastveit et al. (2017). The non-studentized bootstrap overcovers, and increasingly so as κ grows; at $\kappa = 0.5$ its exceedance frequencies for the 5% and 10% quantiles are 0.009 and 0.031. A similar failure of the bootstrap in high dimensions is documented by El Karoui and Purdom (2018) for inference on regression coefficients (see Chernozhukov et al., 2023, for a review). Studentizing the bootstrap with fitted residuals largely solves the problem. The conformal methods are much less affected by the dimension than the non-studentized bootstrap, although they become somewhat conservative as κ increases. A plausible explanation is that the errors collected early in the sample come from models estimated on few observations relative to the number of regressors. These errors are larger than those of the final model, which inflates the empirical quantile. Overall, Table 2 shows that in high-dimensional problems either a studentized bootstrap or conformal quantiles are approximately valid.

Table 2: Monte Carlo simulation: AR(1) with exogenous regressors

	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	$\kappa = 0.10$	$\kappa = 0.25$	$\kappa = 0.50$	$\kappa = 0.10$	$\kappa = 0.25$	$\kappa = 0.50$	$\kappa = 0.10$	$\kappa = 0.25$	$\kappa = 0.50$
CP1	0.011	0.008	0.011	0.046	0.036	0.033	0.090	0.076	0.070
CP2	0.008	0.005	0.003	0.040	0.029	0.024	0.086	0.069	0.060
FpB	0.009	0.004	0.001	0.041	0.027	0.009	0.087	0.065	0.031
FsB	0.012	0.011	0.010	0.051	0.048	0.049	0.099	0.096	0.097
PI-CP	0.984	0.989	0.992	0.921	0.940	0.947	0.831	0.859	0.872
PI-FpB	0.981	0.990	0.998	0.917	0.943	0.978	0.827	0.866	0.930
PI-FsB	0.976	0.977	0.977	0.899	0.900	0.896	0.802	0.802	0.795

Note: Results from 1,000 replications of $y_t = 0.5y_{t-1} + X'_{t-1}\beta + \epsilon_t$ with $\epsilon_t \sim N(0,1)$, each element of the $n \times K$ regressor matrix X drawn from $N(0,1)$ and β a vector of ones. The sample size is $n = 100$ (T in the text) and $\kappa = K/n$ measures the dimension of the problem. Exceedance frequencies are computed from 1,000 draws of y_{n+1} for each replication and averaged across replications. Models as in Table 1.

The studentized bootstrap, however, is sensitive to outliers. Table 3 repeats the AR(1) design with normal errors and $T = 100$, but replaces one error in the sample by a draw from

[4] To ensure that the least squares estimator is unique, we start collecting errors for CP1 in periods 20, 35 and 60, respectively.

$-|N(\mu_o, 0.5^2)|$, so that 1% of the sample is an outlier. Future errors are drawn from the corresponding mixture distribution, so that the sample and the future observations come from the same distribution. CP1 is unaffected by the outlier: its exceedance frequencies hardly change with μ_o and are close to those of the outlier-free design with $T = 100$ in Table 1. The non-studentized bootstrap is also close to the nominal level. The studentized bootstrap, in contrast, overcovers increasingly as the outlier magnitude grows: its exceedance frequency for the 5% quantile falls from 0.048 for $\mu_o = 5$ to 0.019 for $\mu_o = 15$, and that for the 10% quantile from 0.097 to 0.057. The reason is that the studentized bootstrap scales the residuals by their sample standard deviation, which is driven by the outlier, whereas the conformal quantile depends on the errors only through an order statistic: a single outlying error can shift $\hat{Q}_\alpha(\mathcal{E}_t)$ by at most one position in the ordered sample, whatever its magnitude. This finding, together with the failure of the non-studentized bootstrap in high dimensions, is relevant for the nowcasting application, where the GDP growth series contains several large negative observations associated with recessions and the MIDAS model contains many regressors. As shown in Section 4, none of the bootstrap procedures attains coverage close to the nominal level there.

Table 3: Monte Carlo simulation: AR(1) with an outlier

	Standard normal errors with one outlier from $-\lvert N(\mu_o, 0.5)\rvert$								
	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	$\mu_o = 5$	$\mu_o = 10$	$\mu_o = 15$	$\mu_o = 5$	$\mu_o = 10$	$\mu_o = 15$	$\mu_o = 5$	$\mu_o = 10$	$\mu_o = 15$
CP1	0.016	0.015	0.016	0.059	0.058	0.062	0.106	0.108	0.109
CP2	0.018	0.016	0.016	0.052	0.046	0.044	0.097	0.088	0.081
FpB	0.012	0.011	0.011	0.052	0.049	0.049	0.100	0.098	0.098
FsB	0.011	0.010	0.010	0.048	0.032	0.019	0.097	0.078	0.057
PI-CP	0.972	0.974	0.974	0.896	0.900	0.897	0.801	0.803	0.802
PI-FpB	0.975	0.976	0.976	0.898	0.901	0.902	0.801	0.804	0.806
PI-FsB	0.977	0.985	0.988	0.903	0.930	0.958	0.805	0.833	0.871

Note: Results from 1,000 replications of $y_t = 0.5y_{t-1} + \epsilon_t$, where exactly one error in the sample is drawn from the outlier distribution $-|N(\mu_o, 0.5)|$ and the future errors are drawn from the mixture $(1-\omega)N(0,1) + \omega(-|N(\mu_o, 0.5)|)$ with $\omega = 0.01$. The sample size is $n = 100$ (T in the text). Exceedance frequencies are computed from 1,000 draws of y_{n+1} for each replication and averaged across replications. Models as in Table 1.

Finally, to study the joint forecasting of multiple quantiles, we simulate $T = 50$, 100 and 200 observations from the DMQ model of Catania and Luati (2023). We use the same sample

to estimate the DMQ and to construct CP1 forecasts of the nine deciles of the one-step-ahead distribution, based on an AR(4) point forecast and errors collected from period 20 onward.[5] Table 4 reports quantile losses of CP1 relative to the DMQ, averaged over 500 replications, so that values below one favor CP1. Even though the data are generated by the DMQ, CP1 performs similarly to the DMQ in finite samples. Its loss is at most 6.1% higher than that of the DMQ, and for $T = 50$ it is lower for the first and last deciles. For $T = 200$ the difference is largest for the lowest decile (5.3%) and essentially vanishes above the median, a pattern that also appears in the empirical application of Section 5.

Table 4: Monte Carlo simulation: quantile loss for deciles relative to the DMQ

	Deciles								
	$\alpha = 0.10$	$\alpha = 0.20$	$\alpha = 0.30$	$\alpha = 0.40$	$\alpha = 0.50$	$\alpha = 0.60$	$\alpha = 0.70$	$\alpha = 0.80$	$\alpha = 0.90$
$n = 50$	0.964	1.013	1.019	1.032	1.043	1.034	1.043	1.010	0.982
$n = 100$	1.017	1.026	1.011	1.011	1.025	1.039	1.034	1.046	1.061
$n = 200$	1.053	1.035	1.017	0.998	1.002	1.006	1.000	0.991	1.001

Note: Quantile loss of CP1 (Algorithm 1 with an AR(4) point forecast) relative to the DMQ for the one-step-ahead forecast of each decile; values below one favor CP1. Data are generated by the DMQ of Catania and Luati (2023) with parameters, in their notation, $\phi = 0.95$, $\gamma = 0.10$, $\alpha = 0.01$ and $\beta = 0.7$. Samples of $n = 50$, 100 and 200 observations (T in the text) are used for estimation. Results are averages over 500 replications.

4 Nowcasting US GDP tail risk

Nowcasting is the prediction of the present, the very near future and the very recent past. The main challenge is to exploit indicators that are available at a daily, weekly or monthly frequency to predict a quarterly variable such as GDP growth before it is published (see Bańbura et al., 2013, for a review). Following CCM2022, we nowcast the 5% quantile of US GDP growth in each of 15 weeks per quarter: four weeks in each month of the quarter and the first three weeks of the following quarter, before the first estimate of GDP is released in the fourth week.

[5] We forecast deciles rather than percentiles to reduce the computational cost of the DMQ.

4.1 Models

Nowcasting models can be broadly divided into full-system and partial approaches. Full-system approaches specify a joint model for all low- and high-frequency variables, for instance a mixed-frequency VAR (Schorfheide and Song, 2015). Partial approaches estimate a single equation for the low-frequency target in which the high-frequency variables are aligned with the low frequency. With monthly indicators and a quarterly target, for example, each indicator enters with one regressor per month of the quarter, so that even a moderate number of indicators produces many regressors. This parameter proliferation is addressed either by Bayesian shrinkage, as in the Bayesian mixed-frequency models with and without stochastic volatility of Carriero et al. (2015) (BMF-SV and BMF), or by restricting the lag coefficients to a parametric polynomial, as in mixed-data sampling (MIDAS) regressions (Ghysels et al., 2004).

CCM2022 show that the BMF and the BMF-SV produce point nowcasts of similar accuracy, but that the BMF-SV produces better quantile and density nowcasts at a much higher computational cost. This raises the question of whether the gain stems from modeling time-varying volatility or whether any well-calibrated quantile around the same point nowcast would do. Our design answers it by building four quantile nowcasts on the same conditional mean model, whose specification and priors follow CCM2022. The *BMF* reports the 5% quantile of the posterior predictive distribution of the model with constant volatility, the *BMF-SV* that of the model with stochastic volatility. *CP1-BMF* applies Algorithm 1 to the point nowcast of the BMF, the mean of its posterior predictive distribution. *CCP-BMF* applies the conditional conformal quantile nowcast (4) to the same point nowcast. CP1-BMF and CCP-BMF thus replace stochastic volatility by a constant and by a state-dependent conformal shift, respectively. We also report *DCP-BMF*, the one-sided distributional conformal prediction of Chernozhukov et al. (2021a) applied to the posterior predictive distribution of the BMF, which calibrates the whole predictive distribution rather than a shift around the point nowcast.

The second group of models uses a MIDAS regression with many high-frequency regressors

whose polynomial lag parameters are restricted to be common across indicators, estimated by parameter profiling (Ghysels and Qian, 2019). Around its point nowcast we compute CP1-MIDAS and CCP-MIDAS as above, CP2-MIDAS, the lower end of the two-sided conformal interval, and FACI-MIDAS, the FACI algorithm of Gibbs and Candès (2024) wrapped around CP1-MIDAS, which allows for arbitrary distribution shifts. Finally, we report three bootstraps for the MIDAS regression: BOOT-MIDAS, the block wild bootstrap of Aastveit et al. (2017) with blocks equal to the forecast horizon,[6] and the two forward bootstraps of Section 3.

4.2 Data and real-time design

We use the data of CCM2022, organized as a weekly calendar of data releases, and the set of indicators that CCM2022 refer to as the *base macro-finance* set. It contains six monthly non-financial indicators: payroll employment, industrial production, real retail sales (nominal sales deflated by the CPI), housing starts, the manufacturing index of the Institute for Supply Management (ISM) and initial claims for unemployment insurance. It also contains four financial indicators: the Chicago Fed's National Financial Conditions Index (NFCI), the S&P 500 index, the term spread between the 10-year and 1-year Treasury yields (constant maturity), and the credit spread between Moody's Baa corporate yield and the 10-year Treasury yield. Some of these variables are also available weekly. Depending on the week, they enter at the weekly or the monthly frequency, and weekly observations are sometimes aggregated; see Table 2 of CCM2022 for details. Our aim is methodological, and the specifications only need to ensure comparability with CCM2022.

All results are real-time and out-of-sample. At every nowcast date, all models are estimated on the GDP vintage available at that date, following the calendar of CCM2022: in weeks 1–3 GDP growth is observed up to quarter $t-2$ and the models use its second lag, and from week 4 onward it is observed up to $t-1$. GDP growth is measured in annualized percentage points. Nowcast errors are recorded from 1985Q1 onward, and the evaluation sample runs

[6] Aastveit et al. (2017) consider models with a single high-frequency regressor.

from 1989Q4 to 2019Q4 (121 quarters). The outcome is the second available estimate of GDP growth in the quarterly real-time vintages (in practice the third release), taken from the CCM2022 replication files. Algorithm 1 is applied separately for each week of the nowcasting cycle, using the errors of the nowcasts made in the same week of previous quarters and an expanding window ($w = \infty$). An error enters the calibration set only once the outcome used for evaluation has been published. This outcome is released at the end of the quarter that follows the target quarter, so the most recent available error is for two quarters earlier ($\ell = 2$) in weeks 1–11 and for the previous quarter ($\ell = 1$) in weeks 12–15. At the start of the evaluation sample, the calibration set therefore contains only 18 or 19 errors.

For the conditional conformal quantile nowcasts, z_t is the most recent weekly value of the NFCI available at the nowcast date. The NFCI is a weighted average of measures of risk, credit and leverage in money, debt and equity markets and in the banking system, standardized to have mean zero and unit variance; positive values indicate financial conditions tighter than average (Brave and Butters, 2011). Following the release calendar of CCM2022, the NFCI of a week becomes available one week later, so the nowcast in week w uses the NFCI of week $w - 1$, and the nowcast in week 1 uses the NFCI of the last week of the previous quarter.[7] The features are $\phi(z_t) = (1, \max(\text{NFCI}_t, 0))'$, the slope is restricted to be non-positive, and it is estimated once $d/\alpha = 40$ quarters of past errors are available (Remark 6). To increase the number of past errors, the calibration set for week w pools the errors of the nowcasts made in weeks $w - 2, \ldots, w + 2$ of earlier quarters. Errors in adjacent weeks are based on similar information sets, and all pooled errors obey the same publication rule. Pooling raises the number of past errors at the start of the evaluation sample from 18 to 90 in week 5.

The COVID-19 period is excluded from the evaluation and discussed separately.

[7] As CCM2022, we use the weekly financial indicators of their data set, which are not real-time vintages. The NFCI is revised as new data arrive, and its standardization uses the full sample, so the threshold $z = 0$ is not known in real time.

4.3 Evaluation

Let $\hat{q}_t$ denote the 5% quantile nowcast for quarter t in a given week. We report the empirical exceedance frequency $\frac{1}{P}\sum_t \mathbf{1}\{y_t \leq \hat{q}_t\}$ over the $P = 121$ evaluation quarters and test whether it equals the nominal level of 5% with a two-sided t-test. About six exceedances are expected, so these tests have limited power. We therefore also report the quantile score

$$\mathrm{QS}_\alpha(\hat{q}_t, y_t) = \big(\mathbf{1}\{y_t \leq \hat{q}_t\} - \alpha\big)\,(\hat{q}_t - y_t), \tag{6}$$

which is a consistent scoring function for the α-quantile and therefore rewards both calibration and sharpness (Gneiting, 2011). We compare average quantile scores with those of the BMF-SV using the test of Diebold and Mariano (1995), with the long-run variance of the loss differential estimated as in CCM2022 (quadratic spectral kernel with AR(1) prewhitening and the bandwidth of Andrews (1991)). To summarize the 15 weeks, we average the exceedance frequencies and the relative quantile scores across weeks, overall and separately for the first and the last four weeks of the cycle, for the 15 quarters of NBER-dated recessions and the 106 quarters of expansions in the evaluation sample, and for the period since 2000.

4.4 Results

Table 5 summarizes the results across the 15 weeks; Tables 6–9 report them week by week. Three findings stand out. Conformal quantile nowcasts are reasonably well calibrated, whereas the MIDAS bootstrap and the constant-volatility BMF are not. Their accuracy is governed by the point nowcast: around the BMF point nowcast, Algorithm 1 recovers about four fifths of the gain from stochastic volatility, around the MIDAS point nowcast it does not come close. The remaining gap to the BMF-SV is concentrated in recessions, and conditioning on financial conditions closes most of it.

Table 5: Nowcasting US GDP: summary across the 15 weeks

	Coverage		Quantile score relative to BMF-SV						DM test
	Exceed.	Rejected	All	Weeks 1–4	Weeks 12–15	Recessions	Expansions	Since 2000	Worse / better
BMF-SV	0.033	6	–	–	–	–	–	–	–
BMF	0.015	14	1.202	1.144	1.265	0.920	1.264	1.083	12 / 0
BOOT-MIDAS	0.017	13	1.207	1.109	1.364	1.203	1.231	1.102	9 / 0
CP1-MIDAS	0.033	4	1.152	1.102	1.230	1.020	1.188	1.050	8 / 0
CP2-MIDAS	0.019	12	1.168	1.084	1.270	1.112	1.195	1.075	8 / 0
FACI-MIDAS	0.046	1	1.150	1.101	1.231	1.059	1.169	1.088	8 / 0
CCP-MIDAS	0.030	8	1.176	1.121	1.239	1.035	1.215	1.057	7 / 0
CP1-BMF	0.037	4	1.044	1.045	1.039	1.491	0.980	1.052	1 / 0
DCP-BMF	0.061	0	1.021	1.012	1.019	1.642	0.921	1.050	0 / 0
CCP-BMF	0.035	7	1.003	0.984	1.046	1.220	0.989	0.981	0 / 0

Note: Real-time 5% quantile nowcasts, evaluation sample 1989Q4–2019Q4. Exceed.: exceedance frequency averaged over the 15 weeks. Rejected: number of weeks (out of 15) in which the two-sided t-test rejects correct coverage at the 10% level. Quantile score relative to BMF-SV: ratio of the average quantile score (6) of the method to that of the BMF-SV, averaged over the weeks, for all evaluation quarters, weeks 1–4 and 12–15 only, the 15 quarters of NBER-dated recessions, the 106 quarters of expansions, and 2000Q1–2019Q4. Worse / better: number of weeks in which the quantile score is significantly higher / lower than that of the BMF-SV (Diebold–Mariano test, 10% level). BMF-SV and BMF: 5% quantile of the posterior predictive distribution of the Bayesian mixed-frequency model with and without stochastic volatility. BOOT-MIDAS: bootstrap of Aastveit et al. (2017). CP1, CP2, FACI, CCP: Algorithm 1, lower end of the two-sided conformal interval, FACI (Gibbs and Candès, 2024) wrapped around CP1, and conditional conformal quantile nowcast (4) with the NFCI, applied to the MIDAS or the BMF point nowcast. DCP-BMF: one-sided distributional conformal prediction (Chernozhukov et al., 2021a) applied to the BMF.

The predictive distribution of the BMF with constant volatility is far too wide in the left tail: its exceedance frequencies are between 0.8% and 3.3%, and the test rejects correct coverage in 14 of the 15 weeks. BOOT-MIDAS has significantly too few exceedances in 13 of the 15 weeks, as predicted by the simulation evidence, because of the dimension of the MIDAS model, the outliers in the sample, or both.[8] The conformal methods are much closer to the nominal level, although most of them are somewhat conservative. CP1-BMF has exceedance frequencies between 2.5% and 5.0% and is rejected only at the 10% level, in weeks 7–10. CP1-MIDAS is rejected in weeks 1–3, when GDP is observed only up to quarter $t-2$, and in week 10. FACI-MIDAS, which adapts the quantile level online, is rejected only in week 2, at the 10% level; in this application its adaptation mainly removes the conservativeness of CP1-MIDAS early in the quarter. CP2-MIDAS has significantly too few exceedances in 12 of the 15 weeks, which indicates that the MIDAS nowcast errors are asymmetric (Remark 4). DCP-BMF is the only method that is not rejected in any week; its exceedance frequency is slightly above the nominal level in most weeks of the second half

[8]We include many high-frequency regressors in one MIDAS model and restrict the polynomial parameters to be common across variables. In line with the outlier simulations, the two forward bootstraps of Section 3 do not solve the problem: both have too few exceedances in every week except week 5 (Table 10).

of the quarter. CCP-BMF has exceedance frequencies between 1.7% and 5.0% and too few exceedances in 7 weeks, significantly so at the 1% level only in week 7, so conditioning makes the quantile nowcasts slightly more conservative. The BMF-SV is well calibrated in the first half of the quarter but has significantly too few exceedances in weeks 8, 9 and 12–15. Since 2000, none of CP1-MIDAS, CP1-BMF, DCP-BMF and CCP-BMF is rejected in any week.

Table 6: Nowcasting US GDP: exceedance frequencies of 5% quantile nowcasts, odd weeks

	Weeks							
	1	3	5	7	9	11	13	15
BMF-SV	0.041	0.050	0.041	0.033	0.025*	0.033	0.017***	0.025*
BMF	0.033	0.008***	0.025*	0.025*	0.008***	0.008***	0.008***	0.008***
BOOT-MIDAS	0.017***	0.008***	0.050	0.008***	0.008***	0.017***	0.017***	0.017***
CP1-MIDAS	0.017***	0.025*	0.033	0.041	0.033	0.041	0.033	0.033
CP2-MIDAS	0.017***	0.033	0.025*	0.017***	0.008***	0.008***	0.008***	0.017***
FACI-MIDAS	0.033	0.041	0.050	0.050	0.058	0.058	0.050	0.050
CCP-MIDAS	0.017***	0.017***	0.058	0.025*	0.025*	0.041	0.041	0.025*
CP1-BMF	0.033	0.033	0.050	0.025*	0.025*	0.041	0.050	0.041
DCP-BMF	0.050	0.058	0.066	0.066	0.050	0.066	0.066	0.074
CCP-BMF	0.025*	0.025*	0.050	0.017***	0.025*	0.041	0.050	0.041

*Note: Empirical exceedance frequencies of real-time 5% quantile nowcasts for odd weeks, 1989Q4–2019Q4. Models as in Table 5. *, ** and *** denote rejection of the null hypothesis of correct coverage at the 10%, 5% and 1% level, based on a two-sided t-test.*

Table 7: Nowcasting US GDP: exceedance frequencies of 5% quantile nowcasts, even weeks

	Weeks						
	2	4	6	8	10	12	14
BMF-SV	0.033	0.041	0.058	0.025*	0.033	0.025*	0.017***
BMF	0.025*	0.017***	0.025*	0.017***	0.008***	0.008***	0.008***
BOOT-MIDAS	0.008***	0.025*	0.033	0.017***	0.008***	0.017***	0.008***
CP1-MIDAS	0.025*	0.033	0.041	0.041	0.017***	0.050	0.033
CP2-MIDAS	0.041	0.033	0.025*	0.017***	0.008***	0.017***	0.008***
FACI-MIDAS	0.025*	0.033	0.050	0.050	0.041	0.058	0.041
CCP-MIDAS	0.017***	0.017***	0.041	0.033	0.017***	0.041	0.033
CP1-BMF	0.033	0.041	0.050	0.025*	0.025*	0.041	0.050
DCP-BMF	0.033	0.066	0.074	0.058	0.050	0.066	0.066
CCP-BMF	0.033	0.025*	0.050	0.025*	0.025*	0.041	0.050

Note: As Table 6 for even weeks, 1989Q4–2019Q4.

Tables 8 and 9 report quantile scores. The first row contains the average score of the BMF-SV, and all other rows report ratios relative to it, so that values below one indicate a better performance than the BMF-SV. The BMF-SV has a lower quantile score than all MIDAS-based methods in every week. The scores of CP1-MIDAS are between 2% and 33% higher, 15% on average, and significantly so in 8 of the 15 weeks. FACI-MIDAS and CP2-MIDAS perform similarly, and BOOT-MIDAS has the highest scores in most weeks. The gap stems from the point nowcast rather than from the conformal step. By Remark 2, a conformal quantile nowcast is the point nowcast shifted by a slowly changing empirical quantile, so it cannot be more accurate than the point nowcast allows, and the MIDAS point nowcasts have a higher root mean squared error than the BMF point nowcasts in every week, for instance 2.14 against 1.79 in week 5 and 1.73 against 1.46 in week 15.

The models built on the BMF point nowcast allow us to decompose the advantage of stochastic volatility. With constant volatility, the quantile score of the BMF is 11% to 28% higher than that of the BMF-SV, 20% on average, and significantly so in 12 of the 15 weeks; this is the gain from stochastic volatility documented by CCM2022. Replacing the constant-volatility predictive quantile by Algorithm 1 reduces the score by 2% to 21% and leaves a gap of 4% on average. The difference to the BMF-SV is significant only in week 2, at the 5% level. The conformal step thus recovers about four fifths of the gain from stochastic volatility without estimating any volatility process. DCP-BMF, which calibrates the full predictive distribution of the BMF, performs similarly, with an average gap of 2% and no significant difference in any week. The split between recessions and expansions in Table 5 shows where the remaining gap lies. In expansions, CP1-BMF is as accurate as the BMF-SV (relative score 0.980), whereas the BMF with constant volatility is 26% worse: its predictive distribution is too wide in normal times. In recessions, the ranking reverses. The BMF is 8% better than the BMF-SV, and CP1-BMF is 49% worse, because a constant shift around the point nowcast cannot widen when downside risk rises. Stochastic volatility adapts in both directions, and the question is whether an observed indicator can deliver the widening in

recessions without a volatility model.

Table 8: Nowcasting US GDP: quantile scores of 5% quantile nowcasts, odd weeks

	Weeks							
	1	3	5	7	9	11	13	15
BMF-SV	0.213	0.203	0.195	0.172	0.161	0.160	0.158	0.160
BMF	1.110	1.147	1.179*	1.176**	1.237***	1.250***	1.268***	1.263***
BOOT-MIDAS	1.244***	1.029	1.230***	1.079	1.142*	1.314***	1.293***	1.455***
CP1-MIDAS	1.178***	1.093	1.326***	1.042	1.021	1.214***	1.127*	1.275***
CP2-MIDAS	1.160**	1.069	1.285***	1.048	1.138*	1.318***	1.177**	1.326***
FACI-MIDAS	1.150**	1.113	1.321***	1.058	1.017	1.226**	1.161*	1.293***
CCP-MIDAS	1.225***	1.078	1.344***	1.152	1.099	1.229***	1.174**	1.290***
CP1-BMF	1.085	0.991	1.068	0.994	1.062	1.062	1.032	1.049
DCP-BMF	1.055	0.968	1.080	0.976	1.020	1.030	1.039	1.003
CCP-BMF	0.982	0.965	0.864	0.963	1.058	1.075	1.055	1.036

*Note: Average quantile score (6) of real-time 5% quantile nowcasts for odd weeks, 1989Q4–2019Q4. The first row reports the average score of the BMF-SV; all other rows report ratios relative to the BMF-SV, so that values below one (shaded) indicate a lower loss than the BMF-SV. Models as in Table 5. *, ** and *** denote significance at the 10%, 5% and 1% level of a Diebold–Mariano test of equal quantile scores relative to the BMF-SV.*

Table 9: Nowcasting US GDP: quantile scores of 5% quantile nowcasts, even weeks

	Weeks						
	2	4	6	8	10	12	14
BMF-SV	0.201	0.196	0.187	0.167	0.160	0.160	0.158
BMF	1.127*	1.190**	1.137	1.198***	1.223***	1.245***	1.283***
BOOT-MIDAS	1.069	1.092	1.186	1.120	1.142*	1.307***	1.401***
CP1-MIDAS	1.065	1.071	1.244**	1.069	1.042	1.210***	1.309***
CP2-MIDAS	1.050	1.057	1.145	1.073	1.096	1.293***	1.283***
FACI-MIDAS	1.068	1.073	1.194*	1.091	1.007	1.202*	1.271***
CCP-MIDAS	1.079	1.101	1.218	1.113	1.051	1.229***	1.262***
CP1-BMF	1.103**	0.999	1.082	1.025	1.032	1.056	1.019
DCP-BMF	1.029	0.995	1.072	0.993	1.013	1.036	0.999
CCP-BMF	1.018	0.970	0.870	1.036	1.052	1.063	1.031

Note: As Table 8 for even weeks, 1989Q4–2019Q4.

Conditioning on the NFCI delivers most of this widening. CCP-BMF matches the BMF-SV on average, with a relative quantile score of 1.003, and has lower scores than the BMF-SV

in weeks 1 and 3–7, by up to 14% in week 5. It is never significantly different from the BMF-SV in either direction. Relative to CP1-BMF, conditioning narrows the gap to the BMF-SV in recessions from 49% to 22%, at almost no cost in expansions (0.989 instead of 0.980). Since 2000 its score is 2% lower than that of the BMF-SV. Counting over all weeks and recession quarters, the outcome falls below the quantile nowcast 33 times for CCP-BMF, against 40 times for CP1-BMF and 31 times for the BMF-SV.

Figure 1 shows how this happens for week 5. Panel A plots the NFCI available at the nowcast date, Panel B the distance between the point nowcast and the 5% quantile nowcast. The distance of CP1-BMF changes only slowly as past errors accumulate. That of CCP-BMF widens sharply when financial conditions tighten in 2008, to more than 5 percentage points. For 2008Q4, when GDP fell by 6.6%, the week-5 quantile nowcast of CCP-BMF is -6.5%, against -4.0% for CP1-BMF and -4.7% for the BMF-SV. In calm periods, CCP-BMF is *narrower* than CP1-BMF, because its intercept no longer has to cover the stress episodes: between 2010 and 2019, its distance averages 2.85 percentage points in week 5, against 3.32 for CP1-BMF and 2.95 for the BMF-SV. The estimated slopes are economically meaningful. At the end of the sample, an NFCI one unit (about one standard deviation) above its average lowers the 5% quantile nowcast by 1.4 to 1.5 percentage points in weeks 1–3, by 1.1 to 1.3 in weeks 4–7, and by about 0.5 in weeks 10–15. As information about the current quarter accumulates, the point nowcast absorbs much of the downside risk signaled by financial conditions. In weeks 8–15 the estimated slope has the wrong sign before 2009 and is therefore set to zero, so that CCP-BMF coincides with Algorithm 1 applied to the pooled errors for most of the sample. Consistent with this, the whole gain of CCP-BMF over CP1-BMF in quantile scores accrues in weeks 1–7. By period, it accrues in the financial crisis from 2007Q4 to 2009Q2, with a small further gain after 2009; in the 1990s conditioning slightly increases the scores, and between 2000 and 2007 it makes no difference.

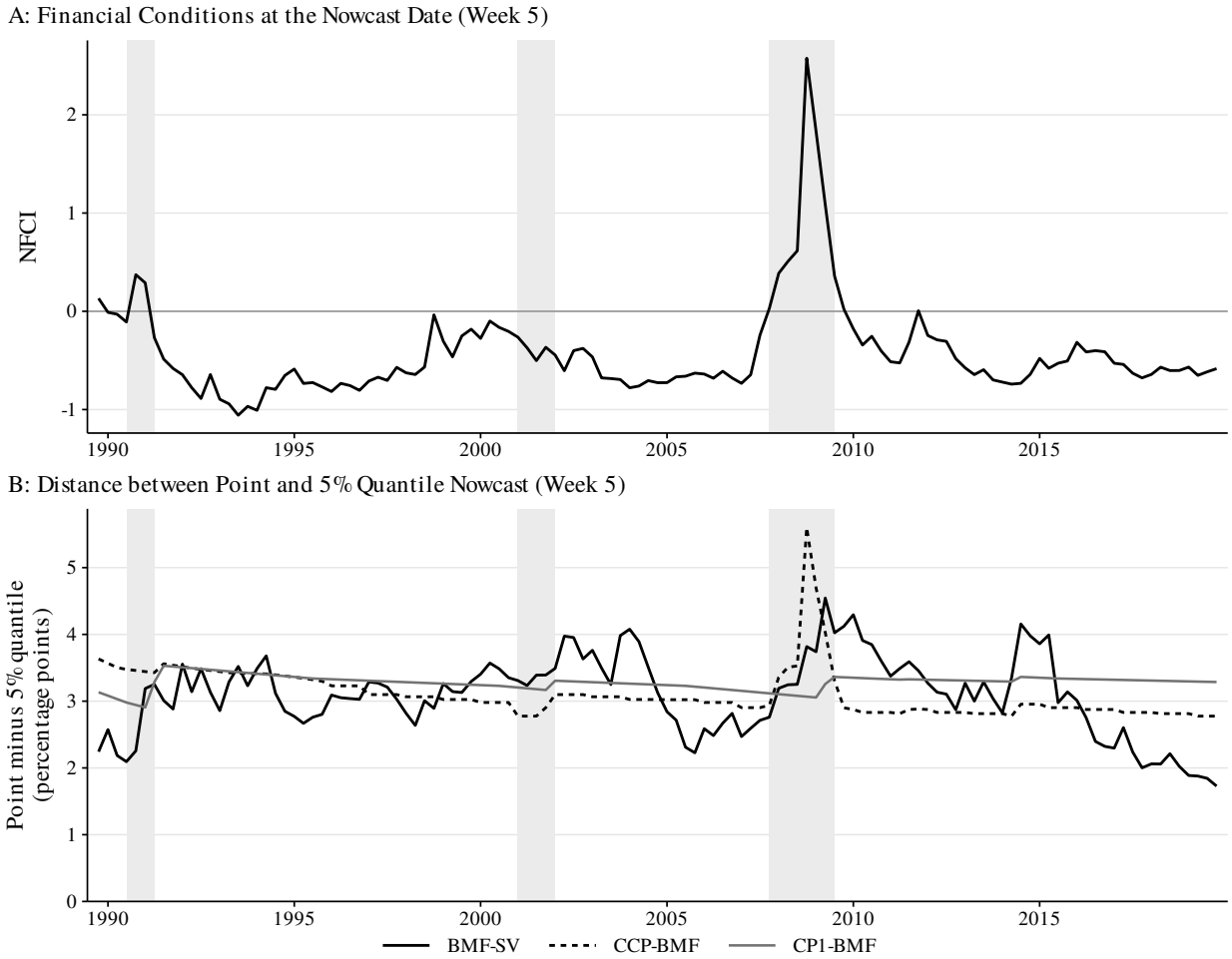


Figure 1: Financial conditions and the downside distance of 5% quantile nowcasts, week 5

Note: Panel A: NFCI available at the week-5 nowcast date; positive values indicate tighter-than-average financial conditions. Panel B: point nowcast minus 5% quantile nowcast for the BMF-SV (posterior predictive mean minus 5% quantile), CP1-BMF and CCP-BMF (BMF point nowcast minus quantile nowcast), in annualized percentage points, 1989Q4–2019Q4. Shaded areas correspond to NBER-dated recessions.

Conditioning does not help the MIDAS nowcasts: CCP-MIDAS has an average relative score of 1.176, against 1.152 for CP1-MIDAS. The MIDAS regression uses the weekly financial indicators directly, and the lower tail of its errors bears no stable relation to the NFCI across weeks. The estimated slope mostly has the wrong sign and is set to zero: it is active in 17% of the MIDAS nowcasts and never in weeks 9–15, against 57% of the BMF nowcasts. When it is active, it does not improve the nowcasts. Conditioning helps only if the indicator predicts the lower tail of the errors of the given point nowcast.

These results speak to the debate on growth-at-risk. In line with Adrian et al. (2019), tight financial conditions signal downside risk beyond the point nowcast of a model that

already contains financial indicators. In line with the caution of Plagborg-Møller et al. (2020), the evidence rests on a single episode, the financial crisis of 2008–2009, and the signal fades within the quarter as real indicators arrive. And in line with Brownlees and Souza (2021), a volatility model and a quantile regression on financial conditions perform equally well on average. Figure 2 shows that they differ in timing. Stochastic volatility widens the quantile nowcasts after large errors have occurred, remaining more pessimistic in 2009 and narrowing in the late 2010s, whereas the conditional conformal quantile nowcast reacts to financial conditions at the nowcast date.

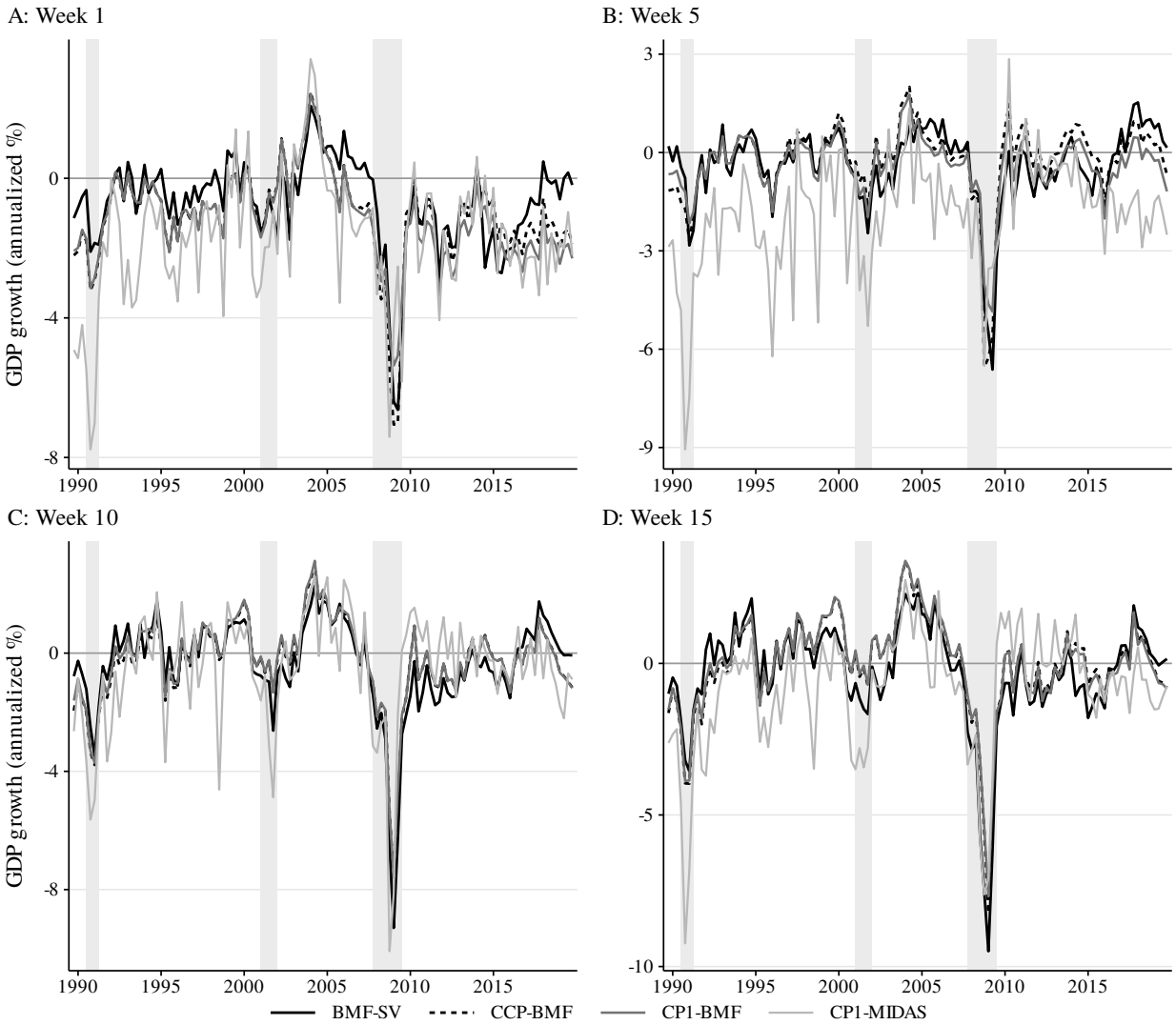


Figure 2: Real-time 5% quantile nowcasts of US GDP growth, 1989Q4–2019Q4

Note: Real-time out-of-sample 5% quantile nowcasts of GDP growth (annualized percentage points) from the BMF-SV, CP1-MIDAS, CP1-BMF and CCP-BMF. Shaded areas correspond to NBER-dated recessions. Panels A–D: weeks 1, 5, 10 and 15.

Table 10 reports the exceedance frequencies of the two forward bootstraps of Section 3 applied to the MIDAS model, together with BOOT-MIDAS. All three bootstraps have too few exceedances in almost every week: the two forward bootstraps are rejected at the 10% level in all weeks except week 5, and BOOT-MIDAS in all weeks except weeks 5 and 6.

Table 10: Nowcasting US GDP: exceedance frequencies of alternative bootstrap procedures

	Weeks							
	1	3	5	7	9	11	13	15
BOOT-MIDAS	0.017***	0.008***	0.050	0.008***	0.008***	0.017***	0.017***	0.017***
FpB	0.017***	0.008***	0.033	0.008***	0.008***	0.008***	0.008***	0.008***
FsB	0.017***	0.025*	0.041	0.017***	0.008***	0.017***	0.008***	0.017***

	Weeks						
	2	4	6	8	10	12	14
BOOT-MIDAS	0.008***	0.025*	0.033	0.017***	0.008***	0.017***	0.008***
FpB	0.008***	0.008***	0.025*	0.008***	0.008***	0.008***	0.008***
FsB	0.017***	0.025*	0.025*	0.017***	0.008***	0.008***	0.008***

Note: Empirical exceedance frequencies of real-time 5% quantile nowcasts for all weeks, 1989Q4–2019Q4. BOOT-MIDAS: bootstrap of Aastveit et al. (2017); FpB: non-studentized forward bootstrap with predictive residuals (Algorithm 3.2 in Pan and Politis (2016)); FsB: studentized forward bootstrap with fitted residuals (Algorithm 3.3 in Pan and Politis (2016)). ∗, ∗∗ and ∗ ∗ ∗ as in Table 6.

4.5 Robustness

Table 11 changes one choice at a time. Computing past errors with the GDP growth of the latest available vintage instead of the evaluation outcome allows a shorter publication lag ($\ell = 2$ in weeks 1–3 and $\ell = 1$ from week 4 onward), at the cost of mixing early and revised releases in the calibration set. For CP1-BMF, this improves calibration (an average exceedance frequency of 4.7% and no rejection) but raises the quantile score by about 4 percentage points relative to the baseline. For CCP-BMF, pooling the errors of adjacent weeks without conditioning does not improve on CP1-BMF (1.058 instead of 1.044), so the gain stems from conditioning. The results are insensitive to the number of quarters required before the slope is estimated (30, 40 or 60) and change little without pooling (1.010) or with the NFCI instead of its positive part (1.019). The sign restriction is essential. Without it, the relative score rises to 1.221, and almost all of the deterioration arises in 2008Q3–2009Q1. Before the crisis, the few episodes of tight financial conditions were not followed by large

negative errors late in the quarter, so the unrestricted slope is positive in weeks 8–15 (between 1.5 and 3.3 in 2008Q2) and *raises* the quantile nowcast when the NFCI spikes. The finite-sample version (5), without sign restriction so that Proposition 2 applies, is very conservative, with an average exceedance frequency of 2.4%, and it is unbounded in 41 nowcasts between 2008Q1 and 2009Q2 (Remark 5).

Table 11: Nowcasting US GDP: robustness of the conformal quantile nowcasts around the BMF

	Coverage		Quantile score relative to BMF-SV						DM test
	Exceed.	Rejected	All	Weeks 1–4	Weeks 12–15	Recessions	Expansions	Since 2000	Worse / better
CP1-BMF	0.037	4	1.044	1.045	1.039	1.491	0.980	1.052	1 / 0
CP1-BMF, errors from the latest vintage	0.047	0	1.082	1.119	1.042	1.610	0.997	1.093	2 / 0
CCP-BMF (baseline)	0.035	7	1.003	0.984	1.046	1.220	0.989	0.981	0 / 0
Unconditional, pooled errors (CP1 with pooling)	0.037	4	1.058	1.076	1.053	1.450	1.003	1.059	2 / 0
NFCI instead of max(NFCI, 0)	0.035	6	1.019	0.982	1.056	1.263	0.999	1.006	0 / 0
No sign restriction	0.042	4	1.221	0.997	1.357	3.123	0.994	1.283	0 / 0
No pooling across weeks	0.036	4	1.010	0.999	1.037	1.353	0.973	1.005	0 / 0
Slope from 30 quarters	0.035	7	1.002	0.984	1.046	1.220	0.989	0.981	0 / 0
Slope from 60 quarters	0.035	7	1.003	0.985	1.046	1.220	0.990	0.981	0 / 0
Finite-sample (augmented) version	0.024	10	–	–	–	–	–	–	–

Note: Columns as in Table 5. First block: CP1-BMF with past errors computed from the evaluation outcome (baseline) and from the latest available vintage. Second block: CCP-BMF with $\phi(z) = (1, \max(\text{NFCI}, 0))'$*, non-positive slope, errors pooled over weeks* $w-2, \ldots, w+2$ *and slope estimated from 40 quarters of past errors on. Third block: one change at a time. The finite-sample version (5) is computed without sign restriction; its lower bound is* $-\infty$ *in 41 nowcasts, so its quantile scores are not defined.*

4.6 COVID-19 and computational cost

Figure 3 extends the sample to the COVID-19 period. Because a conformal quantile nowcast is the point nowcast shifted by a quantile of past errors, it is only as robust as the point nowcast. Fed with the extreme indicator values of 2020, the MIDAS point nowcasts swing far beyond the realized outcomes, for instance to −90% for 2020Q2 in week 5 and to +37% for 2020Q3 in weeks 1 and 5, and CP1-MIDAS follows them. The BMF point nowcasts are much more stable. From week 5 onward, CP1-BMF and CCP-BMF behave much like the BMF-SV: for 2020Q2, when GDP fell by 37.7%, the week-5 quantile nowcasts are −41.9% and −41.4%, against −50.8% for the BMF-SV. In week 1, before any data for the quarter are available, they inherit the rebound extrapolated by the BMF point nowcast for 2020Q3. Conditioning makes little difference in this episode, because policy interventions kept the NFCI low: its highest value at a nowcast date in 2020 is 0.33, and CCP-BMF and CP1-BMF differ by at most 0.8 percentage points. A recession that does not originate in, or quickly

spread to, financial markets is not signaled by financial conditions, as in 2001.

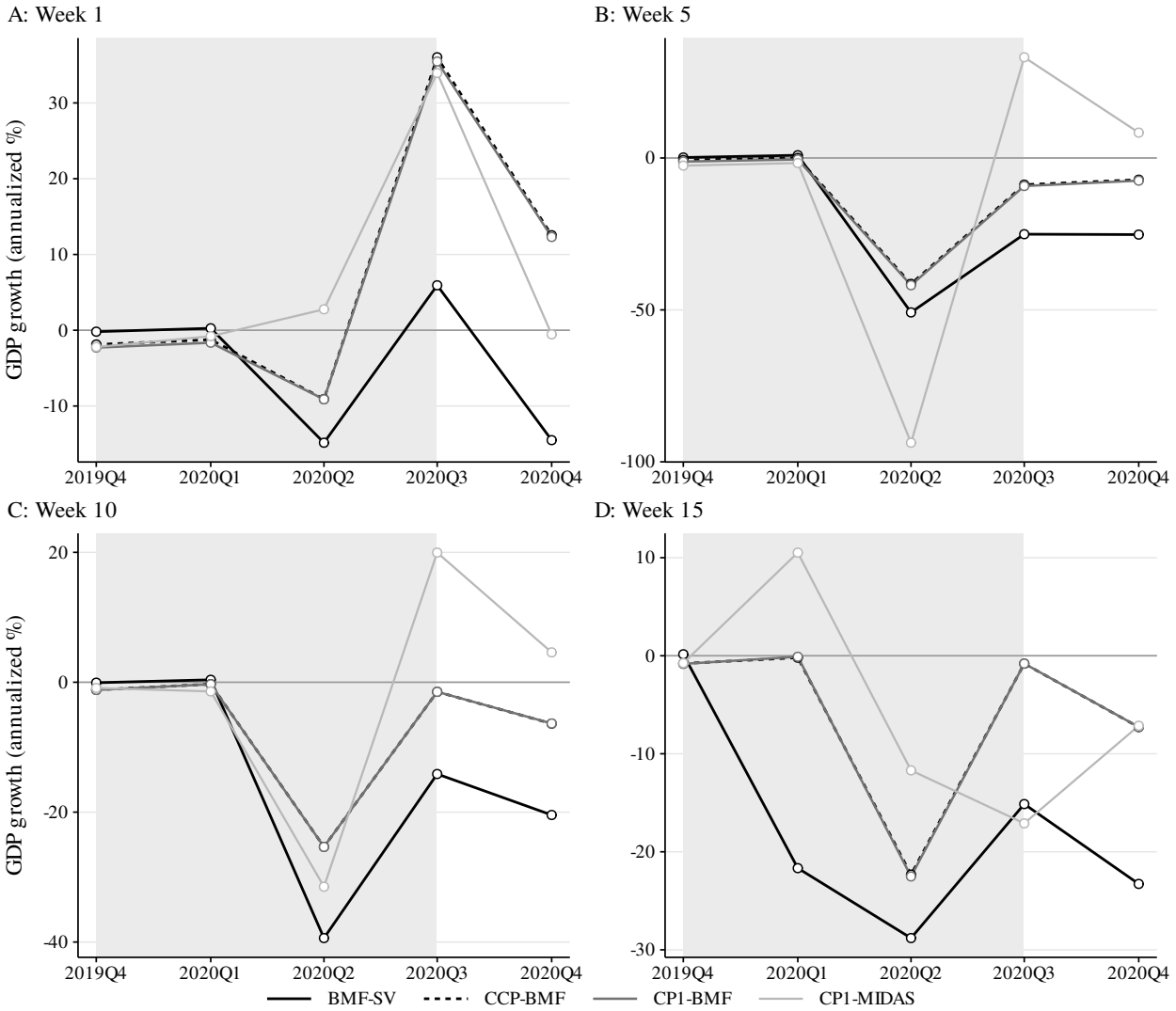


Figure 3: Real-time 5% quantile nowcasts of US GDP growth during COVID-19, 2019Q4–2020Q4

Note: Real-time out-of-sample 5% quantile nowcasts of GDP growth (annualized percentage points) from the BMF-SV, CP1-MIDAS, CP1-BMF and CCP-BMF. Shaded areas correspond to NBER-dated recessions. Panels A–D: weeks 1, 5, 10 and 15.

The computational difference between the approaches is large. On our computing cluster, a single BMF-SV nowcast with the MCMC settings of CCM2022 takes about 17 seconds, against less than one second for the BMF and about 7 seconds for the three MIDAS bootstraps together. Given the point nowcast, CP1 requires sorting the past errors and CCP the solution of one small linear program. Both are negligible. Over the 15 weeks and 144 quarters of the exercise, CCP-BMF thus requires about one twenty-fifth of the computing time of

the BMF-SV. It also extends to any point nowcasting model, including models for which stochastic volatility is not readily available.

5 Forecasting multiple quantiles

When several quantiles are forecast separately, the forecasts may cross, which is incoherent (Bassett and Koenker, 1982). The DMQ model of Catania and Luati (2023) is a recent score-driven model designed to forecast multiple quantiles while avoiding crossing. Its authors show that it outperforms the multiple-quantile CAViaR model (Engle and Manganelli, 2004; White et al., 2015) and ARMA-GARCH models (Bollerslev, 1986). This section compares conformal quantile forecasts, which cannot cross by Proposition 1(iv), with the DMQ in the empirical application of Catania and Luati (2023).

The target is the daily logarithmic return, in percentage points, of Microsoft Corporation stock from December 8, 2010, to November 15, 2018. We forecast all percentiles one day ahead and report only out-of-sample results. As in Catania and Luati (2023), the first 1,000 observations are used only for training. The DMQ is estimated once on these observations, and its quantiles are then updated with the most recent observations while its parameters are kept fixed. For the conformal methods, the point forecast is an AR(4) model re-estimated every day on an expanding window. Algorithm 1 uses a rolling window of the $w = 100$ most recent one-step-ahead errors with $\ell = 1$, that is, the errors of the 100 days preceding the forecast day. We also consider a version of CP1 with scaled scores, in the spirit of the locally weighted conformal prediction of Lei et al. (2018), which we call CP1-EWMA. Each past error is divided by a real-time volatility forecast, $z_s = e_s/\hat{\sigma}_s$, where $\hat{\sigma}_s^2 = \lambda\hat{\sigma}_{s-1}^2 + (1-\lambda)e_{s-1}^2$ is the exponentially weighted moving average of squared past errors with the RiskMetrics value $\lambda = 0.94$, and the forecast is $\hat{y}_t + \hat{\sigma}_t\,\hat{Q}_\alpha(\{z_s\})$. Proposition 1 applies to the scaled scores, and the forecasts of different quantiles still cannot cross, because $\hat{\sigma}_t > 0$. Unlike the conditional procedure of Section 2.3, which would require 99 separate quantile regressions

and could produce crossing quantiles (Remark 7), scaling moves all percentiles proportionally. Finally, we wrap FACI around CP1. Re-estimating the AR(4) every day takes seconds on a laptop, whereas re-estimating the DMQ every day would be prohibitively expensive. The comparison therefore reflects what each method can feasibly do in real time. Because daily returns are nearly unpredictable in mean, CP1 is close in spirit to historical simulation with a rolling window. The comparison therefore also shows how much a carefully specified quantile model gains relative to a simple, model-free benchmark with finite-sample guarantees.

Panels A–D of Figure 4 display the one-step-ahead forecasts of the 5%, 10%, ..., 95% quantiles from CP1, CP1-EWMA, FACI and the DMQ. The DMQ produces smoother trajectories, reflecting its semi-parametric structure. Panels E and F report p-values of the unconditional and conditional backtests of Kupiec (1995) and Christoffersen (1998). Under the null hypothesis, the indicator that the return falls below the forecast α_k-quantile is an i.i.d. Bernoulli variable with success probability α_k. Following Catania and Luati (2023), we conduct the tests separately for each percentile. The unconditional test checks correct coverage on average, and the conditional test additionally checks the independence of exceedances over time. In the unconditional test, all methods perform well. At the 5% level, the null is rejected for 7 of the 99 percentiles for CP1, 5 for CP1-EWMA, one for FACI and none for the DMQ. This is what Proposition 1 and the results of Oliveira et al. (2024) lead us to expect for the conformal methods. The conditional test is rejected more often for CP1 (19 percentiles) than for the DMQ (5 percentiles): the exceedances of CP1 cluster in time. This is the pattern predicted by Remark 2. CP1 targets the marginal quantiles of the errors and is calibrated on average, but it does not adapt to the volatility clustering of daily returns, which the DMQ models explicitly. Scaling the errors by a volatility forecast halves the number of rejections of the conditional test (10 percentiles for CP1-EWMA). FACI adapts to local distribution shifts and achieves the best conditional coverage (one rejection), but, unlike the other methods, it cannot guarantee monotonicity across quantile levels: adjacent FACI percentiles cross in 3,422 of the 98,000 pairs of forecasts.

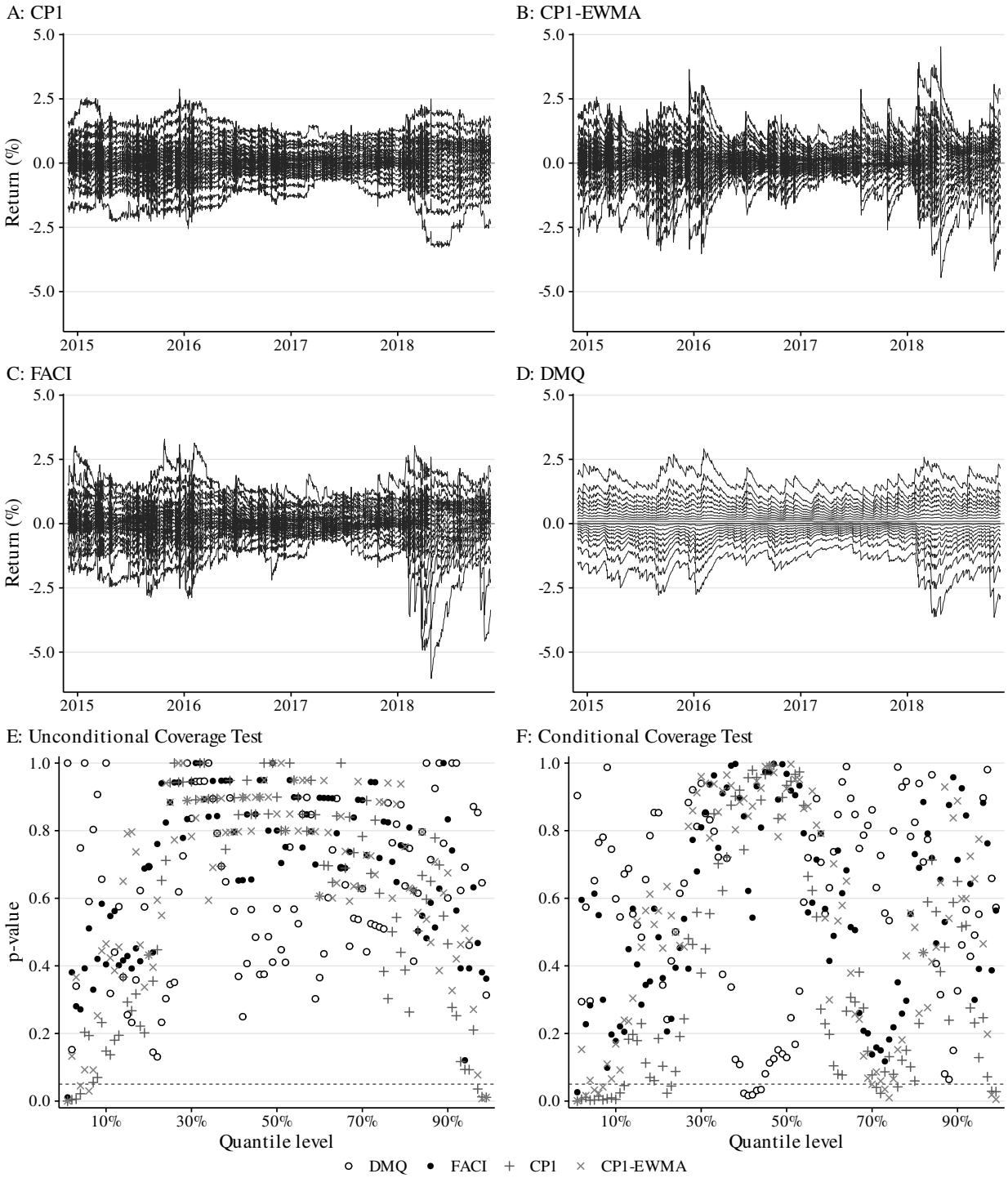


Figure 4: Microsoft returns: quantile forecasts and backtests

Note: Panels A–D: one-step-ahead out-of-sample forecasts of the 5%, 10%, . . . , 95% quantiles of daily Microsoft returns from CP1 (Algorithm 1 with an AR(4) point forecast), CP1-EWMA, FACI wrapped around CP1 and the DMQ. Panels E and F: p-values of the unconditional and conditional backtests of Kupiec (1995) and Christoffersen (1998) for each percentile.

Table 12 reports aggregate quantile losses, defined as the sum of the percentile losses within each decile, relative to the DMQ, so that values below one favor the conformal methods.

The DMQ has the lowest loss in every decile. The losses of CP1 are 0.5% to 0.9% higher in the deciles between the 30th and the 90th percentile and 2.4% and 6.5% higher in the two outer deciles; those of FACI are 1.0% to 5.3% higher. Scaling the errors by a volatility forecast reduces the gap mainly in the tails, to 5.7% in the first and 1.9% in the last decile, and to 0.2% to 0.4% between the 60th and the 90th percentile. The 75% model confidence set of Hansen et al. (2011) contains the DMQ for all deciles, CP1 and CP1-EWMA for all deciles except those around the median (the fifth and sixth), and FACI for the first two deciles. The gap is largest in the tails, where the volatility clustering captured by the DMQ matters most (Remark 2). That the simple volatility scaling closes only part of it suggests that the remaining gap in the outer deciles also reflects the limited resolution of 100 past errors for the extreme percentiles (Remark 3) and the asymmetric response of volatility to negative returns, which the DMQ captures and a symmetric EWMA does not. The simulations in Section 3 show a similar ordering when the data are generated by the DMQ: conformal quantile forecasts are slightly worse than the correctly specified model, but not by much. In this application, the price of a model-free approach that guarantees non-crossing quantiles and requires no estimation beyond an autoregression is a loss that is less than 1% higher in the center of the distribution and up to 6.5% higher in the tails. Estimating the AR(4) only once on the training sample, instead of every day, raises the losses of CP1 by a further 0.3 to 0.8 percentage points.

Table 12: Microsoft returns: aggregate quantile loss relative to the DMQ

	Aggregate percentiles									
	[0.01,0.10)	[0.10,0.20)	[0.20,0.30)	[0.30,0.40)	[0.40,0.50)	[0.50,0.60)	[0.60,0.70)	[0.70,0.80)	[0.80,0.90)	[0.90,0.99]
FACI	1.053†	1.016†	1.017	1.011	1.013	1.014	1.010	1.012	1.013	1.030
CP1	1.065†	1.020†	1.015†	1.007†	1.009	1.008	1.005†	1.007†	1.007†	1.024†
CP1-EWMA	1.057†	1.019†	1.015†	1.009†	1.010	1.007	1.002†	1.004†	1.004†	1.019†
DMQ	1.000†	1.000†	1.000†	1.000†	1.000†	1.000†	1.000†	1.000†	1.000†	1.000†

Note: Aggregate quantile loss of CP1 (Algorithm 1 with an AR(4) point forecast), of CP1-EWMA (CP1 with errors scaled by an EWMA volatility forecast, $\lambda = 0.94$) and of FACI wrapped around CP1, relative to the DMQ, so that values below one indicate a lower loss than the DMQ. The aggregate loss of each decile is the sum of the quantile losses of the percentiles in that decile. †*: the model belongs to the 75% model confidence set of Hansen et al. (2011) (statistic $T_{\max}$, 5,000 bootstrap replications) for that decile; the DMQ row equals one by construction.*

Table 13 and Figure 5 show how the results depend on the window length w. For $w < 100$, some forecast percentiles coincide by construction (Remark 3). A short window of $w = 50$

errors yields the highest losses in every decile and the most rejections of the unconditional test (20 of 99 percentiles). Doubling the window from 100 to 200 errors changes the decile losses by less than half a percentage point, except in the first decile, and further improves the unconditional backtests (5 rejections), but the conditional test is rejected more often (26 rather than 19 percentiles), because a longer window adapts more slowly to changes in volatility. The intermediate value $w = 100$ used above balances these two considerations.

Table 13: Microsoft returns: aggregate quantile loss for different window lengths

	Aggregate percentiles									
	[0.01,0.10)	[0.10,0.20)	[0.20,0.30)	[0.30,0.40)	[0.40,0.50)	[0.50,0.60)	[0.60,0.70)	[0.70,0.80)	[0.80,0.90)	[0.90,0.99]
w=50	1.074	1.036	1.025	1.021	1.025	1.025	1.020	1.019	1.013	1.046
w=100	1.065	1.020	1.015	1.007	1.009	1.008	1.005	1.007	1.007	1.024
w=200	1.051	1.023	1.019	1.011	1.008	1.007	1.005	1.004	1.005	1.028

Note: As Table 12 for CP1 with rolling windows of $w = 50$, 100 and 200 past errors. Percentile losses are aggregated into decile losses.

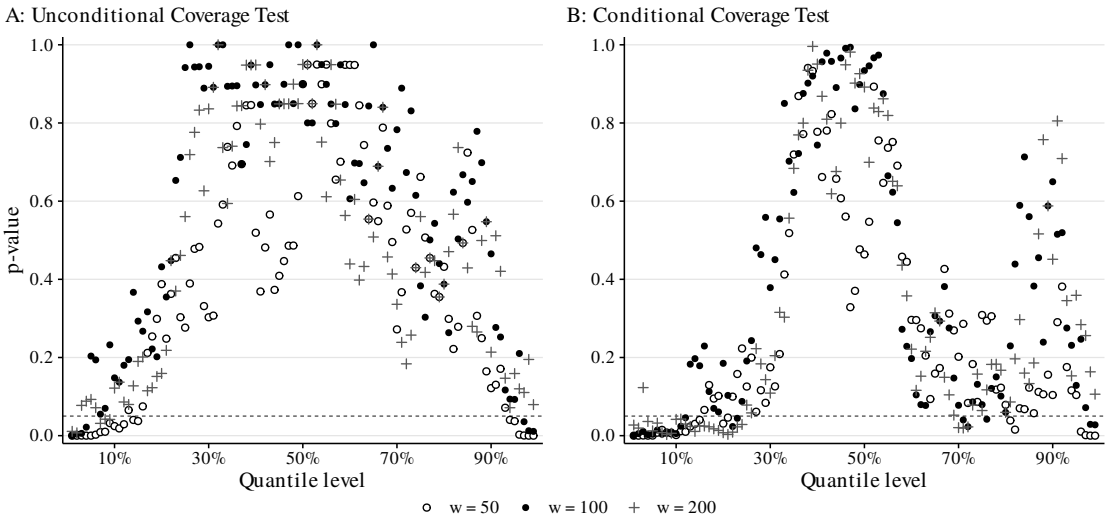


Figure 5: Microsoft returns: backtests for different window lengths

Note: p-values of the unconditional and conditional backtests of Kupiec (1995) and Christoffersen (1998) for CP1 with rolling windows of different lengths.

6 Conclusion

This paper proposes conformal quantile nowcasting, which turns the point nowcasts of any model into quantile nowcasts by adding empirical quantiles of the model's past real-time signed errors. The procedure is a one-sided version of split conformal prediction. It is calibrated in finite samples when the errors are exchangeable and approximately calibrated

for weakly dependent data, quantile nowcasts at different levels cannot cross, and it costs little more than the point nowcast itself. Building on Gibbs et al. (2025), we extend it to a conditional version that replaces the empirical quantile by a quantile regression of past errors on observed risk indicators. In the spirit of the growth-at-risk literature, this lets the lower tail widen when financial conditions tighten.

In simulations, conformal quantile forecasts compare well with residual bootstraps for autoregressions. With many exogenous regressors, the non-studentized bootstrap fails, and the MIDAS bootstrap of Aastveit et al. (2017), which was not designed for this case, is miscalibrated in our application. The studentized bootstrap performs well in high dimensions but is sensitive to outliers, whereas conformal quantiles are not.

In real-time nowcasts of the 5% quantile of US GDP growth, the results decompose the advantage of stochastic volatility. Around the point nowcast of a Bayesian mixed-frequency model, the conformal step recovers about four fifths of the gain from stochastic volatility relative to the same model with constant volatility. The remaining gap arises in recessions, when a constant shift cannot widen. Conditioning on the NFCI closes most of it, so that the conditional conformal quantile nowcast matches the model with stochastic volatility on average, at a small fraction of its computational cost. The gains from financial conditions are concentrated early in the quarter and in the financial crisis of 2008–2009, and they vanish in recessions that financial conditions do not signal, as in 2001 and 2020. Around the less accurate point nowcasts of a MIDAS regression, conformal quantile nowcasts are better calibrated than MIDAS bootstraps but clearly less accurate than the stochastic volatility model. When forecasting all percentiles of daily stock returns, a conformalized autoregression comes close to the DMQ, with the largest differences in the tails, while avoiding quantile crossing by construction.

The approach has limitations. First, a conformal quantile nowcast is only as accurate, and only as robust, as the underlying point nowcast. In the GDP application, the less accurate MIDAS point nowcasts translate into higher quantile scores, and during COVID-19

extreme point nowcasts translate one for one into extreme quantile nowcasts. Second, the unconditional procedure calibrates quantiles marginally, and the conditional procedure adapts only to the indicators it conditions on. Our evidence for financial conditions rests largely on a single episode, which echoes the caution of Plagborg-Møller et al. (2020). Third, with the short samples typical of macroeconomics, the finite-sample guarantees are of limited practical use: the number of past errors limits the quantile levels that can be targeted, and the finite-sample conditional version becomes unbounded precisely when financial conditions reach unprecedented values. The plug-in version with sign restrictions that we use in practice performs well, but it does not inherit these guarantees. Fourth, the guarantees rely on exchangeability, and the results for dependent data are approximate.

Despite these limitations, the results suggest a simple division of labor for quantile nowcasting. The point nowcasting model, which can be as rich as the data allow, delivers the conditional mean. A conformal layer, possibly conditioned on a few risk indicators, delivers calibrated quantiles around it at negligible cost. Natural extensions include other risk indicators, such as recession probabilities or measures of uncertainty, joint forecasts of several quantile levels combined with rearrangement, and forecasts at longer horizons.

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A Proofs

Proof of Proposition 1

Part (i). Let R denote the rank of e_t among the $n+1$ errors in $\mathcal{E}_t \cup \{e_t\}$, so that $R = 1 + |\{s \in S_t : e_s < e_t\}|$, where S_t indexes the errors in $\mathcal{E}_t$. Because the errors are exchangeable and almost surely distinct, every ordering of the $n+1$ errors is equally likely, and R is uniformly distributed on $\{1, \ldots, n+1\}$ (see, e.g., Vovk et al., 2005; Kuchibhotla, 2021). Since $y_t - \hat{y}_t = e_t$,

$$y_t \leq \hat{y}_t + e_{(k)} \iff e_t \leq e_{(k)} \iff |\{s \in S_t : e_s < e_t\}| \leq k-1 \iff R \leq k,$$

where the second equivalence uses that the errors are distinct: $e_t \leq e_{(k)}$ holds if and only if fewer than k elements of $\mathcal{E}_t$ are smaller than e_t. Hence $P(y_t \leq \hat{y}_t + e_{(k)}) = P(R \leq k) = k/(n+1)$.

Part (ii). For the conservative version, $\hat{q}_t(\alpha) = \hat{y}_t + e_{(k)}$ with $k = \lfloor \alpha(n+1) \rfloor$, which satisfies $\alpha(n+1) - 1 < k \leq \alpha(n+1)$ and $k \geq 1$ whenever $\alpha \geq 1/(n+1)$. The result follows from part (i).

Part (iii). For the interpolated version, $\hat{Q}_\alpha(\mathcal{E}_t)$ lies between $e_{(j)}$ and $e_{(j+1)}$, where $j = \lfloor (n-1)\alpha \rfloor + 1 \leq n-1$ for $\alpha < 1$. Hence

$$\{y_t \leq \hat{y}_t + e_{(j)}\} \subseteq \{y_t \leq \hat{q}_t(\alpha)\} \subseteq \{y_t \leq \hat{y}_t + e_{(j+1)}\},$$

and part (i) yields the bounds $j/(n+1)$ and $(j+1)/(n+1)$. Since $(n-1)\alpha < j \leq (n-1)\alpha+1$, the lower bound is at least $\alpha - 2\alpha/(n+1)$ and the upper bound at most $\alpha + 2(1-\alpha)/(n+1)$, which implies $|P(y_t \leq \hat{q}_t(\alpha)) - \alpha| \leq 2/(n+1)$.

Part (iv). Both versions of $\hat{Q}_\alpha(\mathcal{E})$ are non-decreasing in α for any fixed set $\mathcal{E}$, and all quantile levels share the same point nowcast $\hat{y}_t$ and the same set $\mathcal{E}_t$. □

Proof of Proposition 2

The argument follows Gibbs et al. (2025). Index the $n+1$ pairs by $i = 1, \ldots, n+1$, with $i = n+1$ the target period, and write $\phi_i = \phi(z_i)$. Let $\beta^* = \hat{\beta}_t(e_t)$ be the quantile regression fitted with the true error imputed, and let $r_i = e_i - \phi_i'\beta^*$ denote its residuals.

Step 1. Let $E^* = \tilde{q}_t^c(\alpha) - \hat{y}_t = \inf\{E : E \geq \phi_{n+1}'\hat{\beta}_t(E)\}$. If $y_t < \tilde{q}_t^c(\alpha)$, then $e_t < E^*$, so e_t does not belong to the set $\{E : E \geq \phi_{n+1}'\hat{\beta}_t(E)\}$, that is, $e_t < \phi_{n+1}'\beta^*$. Hence $\mathbf{1}\{y_t < \tilde{q}_t^c(\alpha)\} \leq \mathbf{1}\{r_{n+1} < 0\}$.

Step 2. Because β^* minimizes the convex function $\sum_i \rho_\alpha(e_i - \phi_i'\beta)$, zero belongs to its subdifferential: there are numbers ψ_i with $\psi_i = \alpha - \mathbf{1}\{r_i < 0\}$ if $r_i \neq 0$ and $\psi_i \in [\alpha - 1, \alpha]$ if $r_i = 0$, such that $\sum_i \psi_i\phi_i = 0$. For $f(z) = \phi(z)'b \geq 0$, this implies $\sum_i f(z_i)\psi_i = 0$ and therefore

$$\sum_{i=1}^{n+1} f(z_i)\big(\alpha - \mathbf{1}\{r_i < 0\}\big) = \sum_{i:r_i=0} f(z_i)\,(\alpha - \psi_i) \geq 0.$$

Step 3. Because $\hat{\beta}_t(e_t)$ is a symmetric function of the $n+1$ exchangeable pairs, permuting the pairs permutes the residuals, so the pairs $(f(z_i), \mathbf{1}\{r_i < 0\})$ are exchangeable. Hence, using Step 2,

$$\mathbb{E}\big[f(z_t)\,\mathbf{1}\{r_{n+1} < 0\}\big] = \frac{1}{n+1}\,\mathbb{E}\Big[\sum_{i=1}^{n+1} f(z_i)\,\mathbf{1}\{r_i < 0\}\Big] \leq \frac{\alpha}{n+1}\,\mathbb{E}\Big[\sum_{i=1}^{n+1} f(z_i)\Big] = \alpha\,\mathbb{E}[f(z_t)].$$

Combining with Step 1 gives $\mathbb{E}[f(z_t)\mathbf{1}\{y_t < \tilde{q}_t^c(\alpha)\}] \leq \alpha\,\mathbb{E}[f(z_t)]$, and dividing by $\mathbb{E}[f(z_t)] > 0$ yields the result. □